

claude-windows / d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd\subagents\workflows\wf\_d2c10282-601\agent-a5fffddeaf184a077.jsonl`
- SHA-256: `ca3b32be4217936afc21b6090c62f5d5ce29a281fa061b3ac331acb79934e8ba`
- Source modified: `2026-07-04T11:53:03+00:00`
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- project: `wf\_d2c10282-601`
- session_id: `d4cf3fd6-b5bd-4ed1-9b57-88eea69333cd`

Transcript

1. user (2026-07-04T11:48:25.935Z)

You are drafting ONE section of the A13 "alpha launch review" for the KhelSutra badminton-highlight alpha
(repo Khelsutra/badminton-highlight-indexer, engine checkout at
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer; SPA/site repo at
C:/Users/avidu/Projects/khelsutra-guru/khelsutra).
Context: the A1 serving stack is LIVE on Cloud Run asia-southeast1 ? control plane service
'rally-control-plane' (rev 00007, edge-auth ON via Cloudflare Access at api.khelsutra.guru,
4-email
allowlist, maxScale=1, per-instance ephemeral SQLite state), L4 GPU Job 'rally-worker' (WASB-
only alpha:
indexing.skip_ai_handoff=true ? no Gemini in the serving config), storage gs://khelsutra-rally-
serving.
Deploy tooling: deploy/cloudrun/gen_service_yaml.py + README.md + Dockerfile; e2e evidence:
docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md (READ
IT).
Tracker: docs/ALPHA_LAUNCH_READINESS.md (\$7 rows; A10/A11 ship-gate + WASB-C3 provenance are
OWNER-GATED
and still open ? factor that into anything launch-facing).
READ-ONLY: do not modify any file, do not run git writes, do not touch GCP. Windows box,
PowerShell tool.
Ground EVERY claim in a file:line or issue number you actually opened ? no invented provenance.
If you
can't verify something, mark it [unverified]. Your output is a MARKDOWN SECTION consumed by an
orchestrator
assembling the review doc ? return it via the schema, self-contained, concise, evidence-cited.

SECTION: ****Operator runbook**** for the live alpha. Cover, with EXACT commands (verify each
against
deploy/cloudrun/README.md, deploy/cloudrun/gen_service_yaml.py, and the runlogs ? cite):
- deploy/redeploy (manifest generator flow incl. the two gotchas: 'services replace' with an
unchanged
manifest does NOT re-resolve :latest (use services update --image), and a redeploy WIPES the
ephemeral
per-instance SQLite state ? engine issue #471);
- rebuild the image (cloudbuild) + re-pin the L4 job;
- add/remove an alpha tester (Cloudflare Access app allowlist ? team
khelsutra.cloudflareaccess.com);
- edge-auth off/on windows (WHY off is only for smoke tests; restore steps + posture
verification curls);
- health/capabilities checks; where logs live (gcloud logging read filters for the service +
job);
- cost hygiene: what runs idle (min-instances 1 CP ~cheap CPU; L4 job only bills per execution),
teardown pointers (deploy/gcp teardown conventions; GCP VM always-delete rule);

- the cert-renewal watch item (#472): check command + what to do if renewal stalls.

2. user (2026-07-04T11:48:36.351Z)

```
1      # Cloud Run GPU serving ? build / deploy / run (COMPUTE_DECOUPLED_SERVING M6/M7)
2
3      The cloud-serving compute path: a control plane (`CloudRunCompute`,
`backend/compute/cloud_run.py`)
4      dispatches a Cloud Run **Job** execution running this image's worker on an **L4**, then
**bucket-polls**
5      a done-marker for completion (D16). Scale-to-zero (D7) ? no warm pool.
6
7      > **Status:** the dispatch shim + image are **landed + mock-tested** (M6). The **real
build + push +
8      > run on an L4** is the **M7** owner step, within the **$100/?10k** budget (D17) ?
confirm cmd + hourly
9      > cost first, **delete every resource after** ([[gcp-vm-always-delete]]). CI does
**not** build this image.
10
11     ## 0. Prerequisites (owner)
12     - A GCP project with billing on + the Cloud Run, Artifact Registry, and Cloud Build APIs
enabled.
13     - GPU quota for L4 (`GPUS_ALL_REGIONS` ? 1) in the chosen region (asia-south1, else
Singapore ? see
14     `deploy/gcp/README.md`; L4 is often stocked-out, fall back per that runbook).
15     - The WASB weights staged in the bucket (WASB-C3 unresolved ? owner-gated):
`gs://<bucket>/models/wasb_badminton_best.pth.tar`.
16
17     ## 1. Build + push the image (Artifact Registry)
18     ```bash
19     PROJECT=<gcp-project>; REPO=rally
20     # L4 GPU is region-limited ? use asia-southeast1 (Singapore), NOT asia-south1 (Mumbai).
21     REGION=asia-southeast1
22     gcloud artifacts repositories create $REPO --repository-format=docker --location=$REGION
2>/dev/null || true
23     # `gcloud builds submit` has NO -f flag, so a non-root Dockerfile needs a build config:
24     gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project $PROJECT .
25     ```
26     *(A CUDA + micromamba image is large; the first build is ~8?10 min. Budget-watch per
D17.)*
27
28     ## 2. Create the Cloud Run **Job** (GPU, scale-to-zero)
29     The proven shape (first real M7 deploy, 2026-07-03): weights staged in a **same-region
bucket**
30     mounted read-only via GCS FUSE (no weights in the image ? WASB-C3; no init-fetch step
needed).
31     ```bash
32     gcloud run jobs create rally-worker \
33         --image $REGION-docker.pkg.dev/$PROJECT/$REPO/$IMG:latest \
34         --region $REGION --gpu 1 --gpu-type nvidia-l4 --no-gpu-zonal-redundancy \
35         --cpu 8 --memory 32Gi \
36         --max-retries 0 --task-timeout 3600 \
37         --set-env-vars WASB_WEIGHTS=/gcs/models/wasb_badminton_best.pth.tar \
38         --add-volume name=serving,type=cloud-storage,bucket=$SERVING_BUCKET \
39         --add-volume-mount volume=serving,mount-path=/gcs
40     ```
41     - **`--no-gpu-zonal-redundancy` is REQUIRED on the CLI**, not just a cost choice (it
also halves
42     the GPU rate ? right for batch jobs): without the flag `gcloud` asks an interactive
Y/n which,
43     in a non-interactive shell, **hangs forever with zero output and nothing created**. An
empty
44     `gcloud` that "did nothing" ? suspect a hidden prompt.
45     - 8 vCPU / 32 GiB: decode + x264 are CPU-side (NVDEC is a deferred D16 knob), so CPU
headroom is
```

```

46      what keeps the paid GPU busy. `/tmp` is **RAM-backed** ? the pulled input video lives
in memory,
47      so size `--memory` for the largest expected upload (a 5.8 GB GoPro file ? 6 GiB of
tmpfs).
48
49      ## 3. Wire the dispatcher (control plane)
50      The control plane runs with the **cloud-serving** profile (`DEPLOYMENT_PROFILE=cloud-
serving` ?
51      `compute_target=cloud_run`). `get_compute_backend(config)` then builds `CloudRunCompute`
from env:
52      ```bash
53      export DEPLOYMENT_PROFILE=cloud-serving
54      export CLOUD_RUN_JOB=rally-worker CLOUD_RUN_REGION=$REGION
GOOGLE_CLOUD_PROJECT=$PROJECT
55      # A `worker infer` (or the API job path) with a gs:// video now DISPATCHES to the Job
instead of
56      # running in-process; the container runs the SAME worker INLINE (compute_target forced
local_gpu).
57      python -m backend.worker infer --video-uri gs://<bucket>/videos/clip.mp4 --out-uri
gs://<bucket> --segmenter wasb_hybrid
58      ```
59      **The AI handoff needs its key IN THE JOB env** (`GEMINI_API_KEY` ? mount a Secret
Manager secret
60      on the job), or run fully-local with `--set indexing.skip_ai_handoff=true` (candidate
windows
61      emitted as rallies ? the eval-parity mode). With neither, the segmenter used to bail to
a silent
62      0-rally "success" **after** the paid GPU pass; the cloud-serving startup checks now fail
this at
63      boot (`AI_HANDOFF_KEY_MISSING`, rc 2) before the video is even pulled.
64
65      ### 3a. Control-plane SERVICE deploy ? required flags for reliable async jobs
66
67      The API server (`/api/process`, `/api/compile` ? #329) runs its **job worker IN-
PROCESS**: the
68      request returns `202` and a background thread drains the job (GPU detection dispatches
to the Job;
69      the reel stitch runs on the control plane itself). On Cloud Run that background thread
only makes
70      progress while the instance has CPU, so a scale-to-zero **service** deploy MUST set:
71
72      **The canonical deploy path is the committed manifest generator** (#473) ? it bakes the
73      cloud-serving config.json into a startup wrapper, sets every flag below, pins
`maxScale=1`
74      (per-instance SQLite state, issue #471), and deliberately does NOT set `CLOUD_RUN_JOB`
(a
75      reserved runtime env name that gets a manifest rejected; the code default is already
right):
76      ```bash
77      python deploy/cloudrun/gen_service_yaml.py --edge-auth on -o service.yaml
78      gcloud run services replace service.yaml --region $REGION # from PowerShell on Windows
79      ```
80      `--edge-auth off` is only for the pre-CF identity-token smoke/e2e window; redeploy with
`on`
81      before testers touch the service again. The flag-based equivalent (for reference ? a
manifest
82      round-trips the bash wrapper's metacharacters, flags don't on Windows):
83      ```bash
84      gcloud run deploy rally-control-plane --image <image> --region $REGION \
85      --no-cpu-throttling \          # CPU stays allocated between requests ? the drain thread
finishes
86      --min-instances 1 \          # one always-warm instance drains reliably (cheap CPU;
NOT the D7 GPU pool)
87      --cpu 2 --memory 4Gi --port 8080 --allow-unauthenticated \
88      --set-env-vars DEPLOYMENT_PROFILE=cloud-

```

```

serving, CLOUD_RUN_REGION=$REGION, GOOGLE_CLOUD_PROJECT=$PROJECT
89     ``
90     Without `--no-cpu-throttling`, a `/api/compile` returns 202 and then **stalls** ? the
reel never
91     finishes and startup crash-recovery would mark it terminal. `min-instances 1` does NOT
contradict
92     D7 (that was the ~$500/mo warm **GPU** pool); this is a ~1-vCPU CPU instance. Defense-
in-depth is in
93     the code: `Database.recover_stale_jobs()` **re-queues** a compile interrupted by a
restart (idempotent
94     stitch) rather than failing it, so even an unlucky scale-down mid-stitch self-heals on
the next drain.
95     Reserve `--allow-unauthenticated` for the pre-CF-auth smoke test only; lock ingress to
the CF proxy
96     (M9) before inviting testers. The image ships the `auth` extra (PyJWT[crypto]), so
flipping
97     `security.edge_auth_enabled` on needs no derived image ? the M9 Cf-Access JWT verify
works out of
98     the box.
99
100    ## 4. M7 acceptance (the owner runlog)
101    Capture a `DEPLOY_REPORT_cloud-serving_<date>.md` under
102    `docs/COMPUTE_DECOUPLED_SERVING/runlogs/` (mirrors the truly-local template): the exact
commands,
103    rally counts, `gpu_active_seconds`/`wall_seconds`/`bytes_out`, a reel smoke-test
(`ffmpeg -f null -`
104    exit 0 + a 5-frame contact sheet), gotchas, and the **$0 teardown audit**.
105
106    ## Gotchas
107    - The dispatched container **must run inline** ? `CloudRunCompute` forces
`compute_target=local_gpu`
108    + `indexing.detector_impl=native` in the per-execution args so the Job never re-
dispatches itself.
109    - Completion is a **bucket marker** (`<out_uri>/_dispatch/<token>.done`), not the Cloud
Run API status
110    ? so it reuses the storage layer + `execution_policy.poll_until` and survives a
control-plane restart.
111    - A **failed** Job (validation rc 2 / segmenter rc 3) now writes a **failure marker**
carrying the
112    real non-zero rc, so the dispatcher's poll terminates **FAST + accurately** (no 30-min
hang on a bad
113    video). Only a **hung/crashed** container (no marker at all) makes the bounded poll
time out
114    (`ExecutionPolicy.max_wait_sec`); the worker CLI maps that to a clean **rc 4**, and
the API path
115    surfaces it as a `cloud dispatch error` job failure (not a traceback).
116    - libx264 encode (parity, D16) + NVDEC decode-only; the image keeps ffmpeg's default
libx264.
117    - **`GOOGLE_CLOUD_PROJECT` is required** for the real client ?
`GoogleCloudRunJobClient.run_job`
118    fails fast with a clear `RuntimeError` if it is unset/empty (it must resolve a real
`job_path`;
119    an unset project would otherwise build a bogus `projects/None/...` name). Export it
(step 3) before
120    dispatching. The guard runs before the `google-cloud-run` import, so it is exercised
in CI even
121    though the SDK isn't a CI/test dependency.
122    - **Hand-rolled `jobs execute --args` with `gs://` URIs:** gcloud's default comma-split
plus its
123    alternate-delimiter syntax both bite ? repeated `--set` tokens trip the comma parser,
and a
124    `^:^` delimiter splits the `:` inside `gs://`. Use a delimiter that appears nowhere in
the args,
125    e.g. `--args='^@@^~v@@infer@@--video-uri=gs://?'`. (The dispatcher path doesn't have
this

```

```

126     problem ? prefer it; this is for one-off diagnostics.) Container-arg overrides must
include the
127     dispatcher's two forced `--set`s: `deployment.compute_target=local_gpu`,
128     `indexing.detector_impl=native`.
129     - **Git Bash (MSYS) on Windows mangles Unix-looking values** before gcloud sees them:
130     `--add-volume-mount ?mount-path=/gcs` became `C:/Program Files/Git/gcs`, and
131     `--command=/bin/bash` became `C:/Program Files/Git/usr/bin/bash` (the container then
exec-fails).
132     Run gcloud commands with such flags from PowerShell (`MSYS_NO_PATHCONV=1` breaks
gcloud's own
133     wrapper). PowerShell has its own trap: quote any flag value containing commas.
134     - **First execution after an image push** spends minutes in "Container image import in
progress"
135     (visible in `executions describe status.conditions`, NOT in logs) ? that's progress,
not a hang;
136     later executions reuse the imported image.
137

```

3. user (2026-07-04T11:48:37.299Z)

```

1     ""Generate the control-plane Cloud Run *service manifest* (COMPUTE_DECOUPLED_SERVING
M8/M9, #473).
2
3     The A1 alpha deploy drove `rally-control-plane` from a hand-generated YAML that lived in
a
4     session scratchpad ? this commits the generator so the deploy is reproducible from a
clean
5     clone. It emits a Knative service manifest for::
6
7         python deploy/cloudrun/gen_service_yaml.py --edge-auth off -o service.yaml
8         gcloud run services replace service.yaml --region asia-southeast1
9
10    Why a MANIFEST (`services replace`) and not `gcloud run deploy` flags: the container
command is
11    a /bin/bash wrapper carrying shell metacharacters (pipes, redirects, &&) ? on Windows
the gcloud
12    cmd-shim mangles those through `--command/--args`, while a YAML round-trips them
verbatim. Run
13    gcloud from PowerShell, not git-bash (MSYS rewrites `/tmp`-style paths).
14
15    The app is configured through a config.json baked INTO the manifest: the wrapper
base64-decodes
16    it to `/tmp/apphome/config.json` and exports `RALLY_APP_HOME=/tmp/apphome` before
exec'ing the
17    server, so one image serves every config shape (config.json wins over the cloud-serving
preset ?
18    backend/config/loader.py). `--edge-auth` is REQUIRED and explicit: `off` is only for the
19    identity-token smoke/e2e window; redeploy with `on` (the Cf-Access JWT verify, M9)
before the
20    service faces testers again.
21
22    Deliberate manifest choices (learned on the A1 deploy):
23    * `run.googleapis.com/cpu-throttling: "false"` + minScale 1 ? the in-process job
worker drains
24    on a background thread; without always-allocated CPU a 202'd compile stalls (README
$3a).
25    * maxScale DEFAULTS TO 1 ? the jobs/videos DB is per-instance ephemeral SQLite, so a
second
26    instance splits the queue and `/api/jobs/{id}` polls (issue #471, option 1). Raise
it only
27    once state is shared.
28    * `CLOUD_RUN_JOB` is NOT set ? it's a reserved runtime env name and the manifest is
rejected
29    with it; the code default ("rally-worker") is already correct.
30    * No secrets here: bucket/project/aud/team-domain are the same non-secret identifiers

```

```

the
31         committed runlogs carry; the Gemini key stays in Secret Manager (the alpha runs
32         `skip_ai_handoff=true` and never needs it).
33         """
34
35     from __future__ import annotations
36
37     import argparse
38     import base64
39     import json
40     import sys
41
42     DEFAULT_PROJECT = "gen-lang-client-0306026826"
43     DEFAULT_REGION = "asia-southeast1"
44     DEFAULT_SERVICE = "rally-control-plane"
45     DEFAULT_SERVING_BUCKET = "gs://khehsutra-rally-serving"
46     DEFAULT_CF_TEAM_DOMAIN = "https://khehsutra.cloudflareaccess.com"
47     DEFAULT_CF_ACCESS_AUD =
"c980da67ed888fff935fcd913ae1eb4da63f9d65805d6e1c5ca76f92f517082b"
48
49
50     def build_config(args: argparse.Namespace) -> dict:
51         """The app config the wrapper materializes at RALLY_APP_HOME/config.json."""
52         bucket = args.serving_bucket.rstrip("/")
53         return {
54             "deployment": {"profile": "cloud-serving", "compute_target": "cloud_run"},
55             # WASB-only alpha: candidate windows ARE the rallies; no Gemini key in this
service.
56             "indexing": {"skip_ai_handoff": True},
57             "storage": {
58                 "backend": "gcs",
59                 "output_root": bucket,
60                 "input_backend": "gcs",
61                 "input_root": f"{bucket}/videos",
62             },
63             "security": {
64                 "edge_auth_enabled": args.edge_auth == "on",
65                 "cf_team_domain": args.cf_team_domain,
66                 "cf_access_aud": args.cf_access_aud,
67                 "allowed_video_root": "/tmp/apphome/output",
68             },
69         }
70
71
72     def build_yaml(args: argparse.Namespace) -> str:
73         """Render the Knative service manifest for `gcloud run services replace`."""
74         cfg_b64 = base64.b64encode(
75             json.dumps(build_config(args), separators=(",", ":")).encode("utf-8")
76         ).decode("ascii")
77         # mkdir BEFORE the redirect (the redirect can't create the directory), then exec so
the
78         # server is PID 1 and Cloud Run's SIGTERM reaches it directly.
79         wrapper = (
80             "mkdir -p /tmp/apphome && "
81             f"printf %s '{cfg_b64}' | base64 -d > /tmp/apphome/config.json && "
82             "export RALLY_APP_HOME=/tmp/apphome && "
83             "exec python3 -m backend.main --server --host 0.0.0.0 --port 8080"
84         )
85         return f"""\
86 apiVersion: serving.knative.dev/v1
87 kind: Service
88 metadata:
89     name: {args.service}
90     labels:
91         cloud.googleapis.com/location: {args.region}

```

```

92 spec:
93     template:
94         metadata:
95             annotations:
96                 autoscaling.knative.dev/minScale: "{args.min_instances}"
97                 autoscaling.knative.dev/maxScale: "{args.max_instances}"
98                 run.googleapis.com/cpu-throttling: "false"
99         spec:
100             timeoutSeconds: {args.timeout}
101             containers:
102                 - image: {args.image}
103                   command: ["/bin/bash"]
104                   args: ["-c", "{wrapper}"]
105                   ports:
106                       - containerPort: 8080
107                   resources:
108                       limits:
109                           cpu: "{args.cpu}"
110                           memory: {args.memory}
111                   env:
112                       - name: DEPLOYMENT_PROFILE
113                         value: cloud-serving
114                       - name: CLOUD_RUN_REGION
115                         value: {args.region}
116                       - name: GOOGLE_CLOUD_PROJECT
117                         value: {args.project}
118                       - name: RALLY_ALLOWED_HOSTS
119                         value: "*"
120     """
121
122
123 def main(argv: "list[str] | None" = None) -> int:
124     p = argparse.ArgumentParser(description=__doc__.splitlines()[0])
125     p.add_argument(
126         "--edge-auth",
127         choices=("on", "off"),
128         required=True,
129         help="REQUIRED and explicit: 'off' only for the identity-token smoke/e2e window;
130 "
131         "redeploy 'on' (M9 Cf-Access verify) before testers touch it again.",
132     )
133     p.add_argument("--project", default=DEFAULT_PROJECT)
134     p.add_argument("--region", default=DEFAULT_REGION)
135     p.add_argument("--service", default=DEFAULT_SERVICE)
136     p.add_argument(
137         "--image",
138         default=None,
139         help="Container image; default = the project's rally-worker:latest (post-#470 it
140 "
141         "carries cloud-run + auth, so the worker image IS the control-plane image).",
142     )
143     p.add_argument("--serving-bucket", default=DEFAULT_SERVING_BUCKET)
144     p.add_argument("--cf-team-domain", default=DEFAULT_CF_TEAM_DOMAIN)
145     p.add_argument("--cf-access-aud", default=DEFAULT_CF_ACCESS_AUD)
146     p.add_argument("--min-instances", type=int, default=1)
147     p.add_argument(
148         "--max-instances",
149         type=int,
150         default=1,
151         help="Default 1: per-instance SQLite state splits across instances (issue
152 #471).",
153     )
154     p.add_argument("--cpu", type=int, default=2)
155     p.add_argument("--memory", default="4Gi")
156     p.add_argument("--timeout", type=int, default=300)

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154     p.add_argument("-o", "--output", default="-", help="Output path, or '-' for
stdout.")
155     args = p.parse_args(argv)
156     if args.image is None:
157         args.image = f"{args.region}-docker.pkg.dev/{args.project}/rally/rally-
worker:latest"
158
159     text = build_yaml(args)
160     if args.output == "-":
161         sys.stdout.write(text)
162     else:
163         with open(args.output, "w", encoding="utf-8", newline="\n") as f:
164             f.write(text)
165     return 0
166
167
168 if __name__ == "__main__":
169     raise SystemExit(main())
170

```

4. user (2026-07-04T11:48:38.285Z)

```

1     # A1 alpha serving ? post-merge e2e deploy test (cloud-serving)
2
3     **Date:** 2026-07-04 . **Region:** asia-southeast1 . **Project:** gen-lang-
client-0306026826
4     **Scope:** A1 (ALPHA_LAUNCH_READINESS $7) closure ? prove the merged serving fixes
5     (#465 input jail, #467 upload staging, #468 drain concurrency, #469 signed-URL download,
6     #470 auth-in-base-image) plus the committed manifest generator (#474) on a REAL deployed
build,
7     then restore the live edge-auth posture.
8
9     > Companion to `DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md` (PR #464): that report
stood the
10    > stack up and FILED #461/#463/#466; THIS report proves their fixes end-to-end on merged
master and
11    > records the one new bug the run surfaced (#475, fixed by #476 mid-run).
12
13    ## 1. What was deployed
14
15    | Resource | Value |
16    |---|---|
17    | Source | merged `master` ? `195932f` (first build) ? **`3c97117`** (final build, incl.
#476) |
18    | Worker image | `?/rally/rally-worker:latest` ? build 1 `sha256:ed2fe448?` (195932f),
build 2 **`sha256:31475f27?`** (3c97117); the Job re-pinned via `gcloud run jobs update` after
each build |
19    | L4 worker **Job** | `rally-worker` ? unchanged shape (nvidia-l4, 8 vCPU/32Gi, FUSE
`khsutra-rally-serving`?`/gcs`) |
20    | Control-plane **Service** | `rally-control-plane` ? rev **00005** (auth-off, build 1)
? rev **00006** (auth-off, final digest) ? rev **00007** (auth-ON restore, final digest).
Deployed FROM THE BASE IMAGE directly ? no derived PyJWT overlay (**#470 proven**); `maxScale=1`
(#471 option 1) |
21    | Manifest | generated by the **committed** `deploy/cloudrun/gen_service_yaml.py` (#474)
? `--edge-auth off` for the test window, `--edge-auth on` to restore. The scratchpad-artifact
gap (#473) is closed: this deploy used only repo-committed tooling |
22
23    ## 2. Boot / health (rev 00005+, identity-token auth)
24
25    - `GET /api/health` ? `200 {"status":"ok",?}`; `GET /api/capabilities` ? `wasb_hybrid`
present,
26    `gemini_key_present=false` (skip_ai_handoff honored), ffmpeg/ffprobe present.
27    - Before-state captured on the outgoing rev 00004: `/api/process` without a CF JWT ?
**401**
28    `edge auth: missing Cf-Access-Jwt-Assertion header` ? the M9 posture was live as

```


documented.

29

30 **## 3. CUJ on the FINAL image (rev 00006, digest `31475f27?`) ? all green**

31

32 1. ****gs:// process**** ? `POST /api/process {gs://?/MahadevpuraSingles.MP4, wasb_hybrid,
33 **compute=cloud}` ? 202 ? `validating ? dispatching (cloud_run) ? done` ? ****48 play****
34 **segments****

35 2. ****#461 upload staging**** ? chunked upload (`init ? part/0 ? complete`) of a
105,345-byte 720p
36 clip ? `POST /api/process` with the returned ****local**** path + `compute=cloud` ? the
job staged
37 it to `gs://khelsutra-rally-serving/videos/d8335f51?/upload-e2e-720p.mp4` (byte-exact
size
38 verified in the bucket) and dispatched to the L4 ? ****success**** (0 segments ? correct
for a
39 synthetic test pattern). NOT the pre-fix "cloud-serving needs a cloud input" hard-
fail.
40 Verified on BOTH builds (`ed2fe448?` and the final `31475f27?`).
41 - ? Upload validation floor: a 640x360 test clip is rejected (`resolution\_fps\_check`,
42 min 1280x720) ? the first attempt failed on this; alpha docs should mention the
floor.
43 3. ****#468 drain concurrency, observed live**** ? the upload job staged + dispatched
****while**** the
44 48-segment reel compile was stitching on the CP: remote dispatch overlapped the local
compile
45 lane exactly as designed (no serialization, no false-empty stall). On rev 00006 the
two process
46 jobs (gs:// + upload) also ran the drain concurrently.
47 4. ****compile**** ? `POST /api/compile {video\_id, segment\_ids:[1..48], "e2e.mp4"}` ? 202 ?
done;
48 measured stitch wall on rev 00005: created 10:39:32 ? finished 10:44:26 (****~4m54s****,
incl. the
49 gs:// source re-fetch); rev 00006 re-run equally green. Reel ****mirrored to**
50 `gs://khelsutra-rally-serving/highlights/915473?/e2e.mp4` ? 174,046,228 bytes (~166
MiB)****.**
51 5. ****#463 signed-URL download ? VERIFIED****:
52 - `GET /api/highlights/915473?/e2e.mp4` ? ****`307`**** ?
53 `https://storage.googleapis.com/khelsutra-rally-
serving/highlights/?/e2e.mp4?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-
Credential=492532266377-compute@?&X-Goog-Expires=3600&X-Goog-Signature=?`
54 (v4, signed AS the runtime SA via keyless signBlob ? no key file anywhere).
55 - `curl -L` through the redirect: ****full 174,046,228 bytes in 8.1s**** (single GET ? no
32 MiB
56 edge-500, no Range workaround), and the file ****ffprobes valid****: H.264 1920x1080 @
29.97 fps
57 + AAC, duration 244.29s (the 48-rally reel).
58

59 **## 4. Bug surfaced mid-run: #475 ? fixed by #476 (merged, redeployed)**

60

61 On the FIRST attempt (rev 00005, build `ed2fe448?` = pre-#476), the #463 check
****failed****:
62 `GET /api/highlights/...` ? ****500****; log:
63 `[highlights] signed-url path failed ?: you need a private key to sign credentials`.
64

65 - The reel WAS mirrored correctly; the ****keyless signBlob path**** failed and the handler
fell back
66 to serving the 166 MiB file locally ? Cloud Run's ~32 MiB cap ? edge-500 (the original
#463
67 symptom, resurrected by a different root cause).
68 - ****Root cause (verified live):**** `GCSStorage.signed\_url()` reused the storage client's
own token
69 for `iamcredentials.signBlob`; `google.cloud.storage.Client.SCOPE` is ****devstorage-**
only******, which
70 signBlob rejects (`ACCESS\_TOKEN\_SCOPE\_INSUFFICIENT`) ? and the branch's bare `except:

```

pass`
71     swallowed the real error, so the fallthrough raised the misleading private-key
message.
72     Environment was clean: the SA holds self-`serviceAccountTokenCreator` and
73     `iamcredentials.googleapis.com` is enabled ? code bug, not config.
74     - **Why tests never saw it:** the fakes carry no `_credentials`; this was the FIRST run
of #469's
75     path on real compute credentials. Filed **475** ? fixed by **476**
(`_signblob_identity`)
76     mints a fresh `cloud-platform`-scoped ADC credential; failures now LOGGED before
falling back;
77     regression tests reproduce the credential shape). Suite 1659? / cov 85.46% / ruff+mypy
clean;
78     merged on green CI per the standing agreement; rebuilt (`31475f27?`) and redeployed
(rev 00006);
79     $3.5 is the pass on the fixed build.
80
81     ## 5. Notes / observations (rough edges captured)
82
83     - **`services replace` with an unchanged manifest is a NO-OP** ? it does not re-resolve
`:latest`
84     after a rebuild. Force a new revision (`gcloud run services update --image ?:latest`)
or stamp a
85     revision suffix. Follow-up candidate: `--revision-suffix` in `gen_service_yaml.py`.
86     - **A redeploy wipes the per-instance SQLite state** ? `/api/highlights` then 404s on
the DB
87     record check (`backend/main.py:1052`) even though the reel mirror exists in gs://.
Known, filed
88     as **471**; the CUJ was re-run on rev 00006 to regenerate state. Alpha-operator
implication: a
89     deploy invalidates in-flight jobs and past download URLs (re-process regenerates
them).
90     - Upload validation floor (?1280x720) ? $3.2.
91     - 422-before-401 quirk: a **malformed-JSON** body gets FastAPI's 422 before the edge-
auth
92     dependency runs; any well-formed request without a CF JWT still 401s. No exposure;
noted so a
93     422 in edge-auth triage isn't misread.
94
95     ## 6. Edge-auth restore + live posture (rev 00007, digest `31475f27?`)
96
97     - Redeployed with the generator's `--edge-auth on` manifest ? rev **00007**.
98     - `GET /api/health` ? **200** (deliberately exempt).
99     - `POST /api/process` (valid JSON, no CF JWT) ? **401**
100     `edge auth: missing Cf-Access-Jwt-Assertion header`.
101     - `https://api.khelsutra.guru/api/health` (no CF session) ? **302** ?
102     `https://khelsutra.cloudflareaccess.com/cdn-cgi/access/login/api.khelsutra.guru??`
with the
103     Access-app aud `c980da67?082b` in the login meta ? the exact pre-test posture,
restored.
104
105     ## 7. A1 DoD checklist
106
107     | DoD item | Status |
108     |---|---|
109     | Serving endpoint deployed + healthy | ? rev 00007, final digest `31475f27?` |
110     | Upload ? process ? jobs ? compile ? download CUJ | ? end-to-end on the final image |
111     | Edge-auth (M9) restored + verified | ? 401 sans JWT; api.khelsutra.guru ? CF login 302
|
112     | >32 MiB download via signed URL (#463) | ? 307 + full 174 MB single-GET + ffprobe-
valid |
113     | Browser-upload staging to gs:// (#461) | ? staged byte-exact + dispatched + success |
114     | Parallel drain (#466/#468) observed live | ? dispatch overlapped compile; 2 process
jobs drained concurrently |
115

```

116 ****A1 ? DONE**** (flipped in `docs/ALPHA\_LAUNCH\_READINESS.md` §7 in this PR).
117

5. user (2026-07-04T11:48:49.844Z)

```
1      # Alpha Launch Readiness Plan
2
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**
2026-07-03; **Last updated:** 2026-07-04
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed
in `docs/README.md` and `docs/DOC_STATUS.md`.
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from
`NEXT_STEPS.md`, `ROADMAP.md`, `docs/data_pipeline/GOLDEN_VIDEOS.md`,
`R2_EVAL_METRIC_DESIGN.md`, `COMPUTE_DECOUPLED_SERVING/`, and the sibling `kheIsutra` Studio
docs.
7      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9      ---
10
11     ## 0. TL;DR
12
13     The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` A2-A9 made the local manifest/path, full-
fusion/audio, decision, and shadow-training surfaces explicit; `[analysis]` A4/A5 refreshed the
n=15 nightly and heuristic floors; `[analysis]` A8/A9 still reject served Gate A default
promotion and owned-model serving claims on the current evidence.
14
15     The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17     ## 1. Problem & goal
18
19     The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
20
21     - **Can an invited tester use the product without SSH?**
22     - **Can the 15-video golden corpus now support honest quality gates?**
23     - **Can we claim or ship an owned/commercial rally detector?**
24
25     This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
reproducible, and the remaining model/legal gates are explicit rather than accidentally implied
by launch activity.
26
27     ## 2. Decisions locked
28
29     | # | Decision | Source / date | Implication |
30     |---|---|---|---|
31     | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32     | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
33     | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md`; `[analysis]` local eval runs, 2026-07-03 |
Rebase/portability, feature completeness, baseline refresh, and gate ratification come before
quality claims. |
34     | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
```

```

`[analysis]` `backend.eval.serve_contrast` run, 2026-07-04 | Proposal-side lift does not survive
served-windowing safety; keep it as an opt-in/analysis lever until owner ratification. |
35 | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36 | D6 | **Gen-0 n=15 shadow training is non-serving.** | `[analysis]`
`training.gen0.harness` run, 2026-07-04 | Learned mean beats the heuristic floor, but the paired
sign test is not significant; #172 and WASB C3 remain hard blockers. |
37
38 ## 3. Verified snapshot, 2026-07-03
39
40 All five Git checkouts were fast-forwarded before analysis:
41
42 | Repo | Branch | Synced HEAD |
43 |---|---|---|
44 | `badminton-highlight-indexer` | `master` | `e4914d1` |
45 | `khelsutra` | `master` | `b0f70a5` |
46 | `rally-annotator` | `main` | `ef9e088` |
47 | `sports-data-collector` | `master` | `bfc014f` |
48 | `sports-obsreport` | `main` | `a0c7252` |
49
50 Corpus/eval facts:
51
52 - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
53 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
54 - `[analysis]` A7 generated full shuttle/person fusion schema for all 15 videos and A8
audio augmentation populated audio features for all 15.
55 - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` now runs
over all 15 and measures person-fusion `Delta F1 = +0.008`; the paired CI crosses zero, so
person fusion is not promotable by itself.
56 - `[analysis]` `python -m backend.eval.fusion_audio --features-dir output --compare`
measures audio `Delta F1 = +0.018`; the paired CI crosses zero, so audio is a shadow/suggestive
cue, not a default-flip.
57 - `[analysis]` `python -m backend.eval.serve_contrast --manifest
output\human_lovo_manifest.json` measured proposal Gate A mean `Delta F1 +0.0287`, but served
mean `Delta F1 -0.0328`; not served-safe.
58 - `[analysis]` `python -m training.gen0.harness --manifest
output\human_lovo_manifest.json --floor eval_baselines\heuristic_lovo_n15.json --version
0.1.1-n15-shadow.20260704 --out artifacts\rally_tal_gen0` produced learned LOVO mean F1 `0.551`,
floor passed versus heuristic `0.446`, but sign test `9/15`, `p = 0.3036`, `ship=False`.
59 - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
60 - `[analysis]` Before A6, `python -m backend.eval.ablation --features-dir output
--granularity group` failed at import due to a circular import between
`backend/eval/ablation.py` and `backend/eval/ablation_pdf.py`; A6 fixes the module entrypoint
and keeps full-fusion ablation scoped to the 6 videos with full schema until A7.
61
62 ## 4. Launch strategy
63
64 ### 4.1 Alpha lane: make the product usable
65
66 The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
67
68 ### 4.2 Quality lane: make the 15-video corpus gate-ready
69
70 The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.

```

```

71
72     ### 4.3 Flywheel lane: make tester feedback useful
73
74     The flywheel lane is successful when user corrections can flow into the collector/vault
75     path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
76     but it should not block the first controlled alpha if manual promotion can bridge the gap.
77
78     ## 5. Risks and guardrails
79
80     | Risk | Why it matters | Guardrail |
81     |---|---|---|
82     | Launch activity implies model launch approval | The engine docs explicitly say rally-
83     detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
84     scoped to invite-only processing, not owned-model claims |
85     | Stale local manifests create false confidence | Manifest-driven tools skipped or
86     failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
87     check |
88     | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
89     n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
90     | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
91     reported stale baselines | Refresh baselines only after reviewer sign-off and record the
92     command/output |
93     | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
94     can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
95     lane is already stable |
96     | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
97     model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
98     launch tasks |
99
100    ## 6. Honest limits
101
102    Completing this doc's alpha lane does **not** prove the owned model is ready, that WASB
103    weights are commercially clean, that the product can serve unbounded public traffic, that multi-
104    sport is ready, or that every tester correction automatically improves the model.
105
106    Completing the quality lane does **not** by itself deploy a service. It only makes the
107    measurement surface honest enough for a launch decision.
108
109    ## 7. Deliverables & progress tracker
110
111    Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
112    shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
113    row explicitly names a cross-repo batch.
114
115    | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
116    |---|-----|-----|-----|-----|-----|---|
117    | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
118    indexer` | - | No | DONE | #451 |
119    | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
120    edge auth ON, upload -> process -> jobs -> compile -> download verified | `khesutra` +
121    `badminton-highlight-indexer` | A0 | Owner spend / CF config | DONE |
122    #462/#464/#465/#467/#468/#469/#470/#474/#476; e2e runlog
123    `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md` |
124    | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
125    check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
126    | A0 | No | DONE | #452 |
127    | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
128    reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
129    collector` + vault | A2 | Owner upload/storage scope | DONE | collector #62 |
130    | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
131    command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
132    | Reviewer sign-off | DONE | #453 |
133    | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
134    floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | DONE | #454 |
135    | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/toIF1 ablation

```

```

on the n=15 corpus | `badminton-highlight-indexer` | A2 | No | DONE | #455 |
106 | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | DONE | #458 |
107 | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | DONE | #458 |
108 | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | DONE | #458 |
109 | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
110 | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
111 | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train`/`reeval` job, delta-F1 surface, and alignment with
`COMPUTE_DECOUPLED_SERVING` M11 | `kheIsutra` + `badminton-highlight-indexer` + collector | A1,
A3 | Owner D13 reopen (`kheIsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) | GATED | - |
112 | A13 | Alpha launch review: runbook, privacy/consent copy, observability/job-failure
triage, known limits, rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11
| Owner launch decision | TODO | - |
113
114 ## 8. Concrete next PR sequence
115
116 1. **Docs PR:** land this tracker.
117 2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
118 3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
119 4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
120 5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
121 6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
122
123 A2 smoke command:
124
125 ```powershell
126 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15
127 python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-
count 15 --write output/human_lovo_manifest.local.json
128 ```
129
130 The checker resolves stale absolute paths against the repo's sibling `Annotation Setup`
folders and `output/`, then exits non-zero if any required `trajectory` or `labels` file is
absent.
131
132 A4 nightly baseline refresh record:
133
134 ```powershell
135 python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --update-baseline
136 ```
137
138 Result:
139
140 ```text
141 INFO baseline snapshot written to C:\Users\avidu\Projects\kheIsutra-guru\badminton-
highlight-indexer\eval_baselines\nightly_baseline.json
142 INFO default: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135
143 INFO milestone: n=15 tolF1=0.364 f1@0.5=0.472 merge_rate=0.135

```

```

144     INFO    baseline_off: n=15 tolF1=0.138 f1@0.5=0.237 merge_rate=0.638
145     ```
146
147     Pass-mode proof:
148
149     ```powershell
150     python scripts\nightly_regression.py --manifest output\human_lovo_manifest.json
--features-dir output --no-report
151     ```
152
153     Result: `PASS`, exit code `0`, baseline snapshot `2026-07-03`, default/milestone `tol_f1
= 0.3636`, `f1@0.5 = 0.4723`, `merge_rate = 0.1350`; baseline-off `tol_f1 = 0.1382`, `f1@0.5 =
0.2366`, `merge_rate = 0.6380`.
154
155     A5 heuristic floor refresh record:
156
157     ```powershell
158     python -m backend.eval.calibrate_local --manifest output\human_lovo_manifest.json
--metric f1 --floor-out eval_baselines\heuristic_lovo_n15.json
159     ```
160
161     Result:
162
163     ```text
164     baseline (untuned) mean held-out f1: 0.237
165     calibrated mean held-out f1:          0.446 +/- 0.239 (gap +0.000)
166     ```
167
168     The committed floor is `eval_baselines/heuristic_lovo_n15.json`: `n = 15`, `mean_f1 =
0.4456`, `std = 0.2390`, `metric = f1`, `iou = 0.5`. Every leave-one-video-out fold selected
`(0.05, 1.0, 2.0, 0)`.
169
170     A6 ablation/R2 record:
171
172     ```powershell
173     python -m backend.eval.ablation --features-dir output --granularity group
174     python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json
--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served
--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\a6_r2_ablation.json
175     ```
176
177     Result: the ablation CLI now runs, but full-fusion feature ablation still reports `n=6`
because only the original six feature JSONs carry the full fusion schema; A7 must
regenerate/upconvert the newer nine before fusion/person/audio promotion decisions. The n=15
R2/tolF1 contrast is positive for the served milestone versus baseline-off: baseline/off `tolF1
= 0.138`, served `tolF1 = 0.364`, `Delta tolF1 = +0.225` with CI `[+0.165, +0.283]`, sign
`14/0`, `p = 0.000`; `Delta F1@0.5 = +0.236` with CI `[+0.161, +0.313]`; merge rate improves
from `0.638` to `0.135`. Net over-segmentation guard passes (`1.276 -> 0.378`), while the strict
segment-count rule remains blocked by increased splits, so R2/tolF1 is the supported gate for
the baseline-off to served milestone contrast rather than a blanket fusion/default-flip
approval.
178
179     A7 full-fusion/audio feature refresh record:
180
181     ```powershell
182     python -m backend.eval.fusion_golden --manifest output\human_lovo_manifest.json --out-
dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json --missing-
fusion-only --smallest-first --max-gpu-temp-c 87
183     python -m backend.eval.fusion_audio --manifest output\human_lovo_manifest.json
--features-dir output --source-manifest ..\sports-data-collector\data\golden_source_videos.json
--augment
184     ```
185
186     Result: the GPU pass completed without a thermal abort after fans were set to max
performance. Full-fusion schema is now present for `15/15` feature JSONs, and audio features are

```

present for `15/15`. The large regenerated clips completed as follows: `mahadevpura\_1` `1` window / `0` positives / `10` GT, `GX020094` `99` / `30` / `40`, `GX030094` `79` / `21` / `41`, `GX010141` `26` / `11` / `29`, `GX010137` `56` / `31` / `34`, `Boxhill\_Doubles` `59` / `12` / `43`, `mahadevpura\_2` `14` / `2` / `40`, `GX010128` `51` / `15` / `52`, `Badminton\_BXH\_2` `29` / `1` / `44`.

187

188 A3 source-manifest follow-up: `sports-data-collector` PR #62 is merged. The authoritative collector manifest now records `13/15` present local source/proxy paths and two Boxhill run-log-only eval inputs: `GX010141\_1080p.mp4` (`daaf8af198640516d14b217f50445d99`) and `GX010137\_1080p.mp4` (`74eedbc35d739c231344b8929df4b4a7`). The F-drive 4K originals are preserved there only as `original\_` provenance, not as eval join targets, because the golden trajectories/labels are 1920-wide. Freshly re-extracting those two Boxhill full-fusion files requires recovering the exact 1080p eval inputs or adding a coordinate-normalization fix; do not substitute the 4K originals for person-cue extraction.

189

190 A8 fusion/default-decision record:

191

192 ```powershell

193 python -m backend.eval.fusion_compare --features-dir output

194 python -m backend.eval.fusion_audio --features-dir output --compare

195 python -m backend.eval.ablation --features-dir output --granularity group

196 python -m backend.eval.serve_contrast --manifest output\human_lovo_manifest.json

197 python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json

--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served

--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\A6_r2_ablation.json

198 ```

199

200 Result:

201

202 - Person fusion is ****not promoted****: shuttle-only F1 `0.302`, shuttle+person F1 `0.311`, Delta F1 +0.008, paired CI `[-0.040, +0.055]`, sign `8W/6L`, `p = 0.791`.

203 - Audio is ****shadow/suggestive, not promoted****: shuttle+person F1 `0.311`, shuttle+person+audio F1 `0.329`, Delta F1 +0.018, paired CI `[-0.017, +0.053]`, sign `10W/4L`, `p = 0.180`.

204 - Group ablation remains non-significant for cue promotion: audio +0.018 CI `[-0.017, +0.053]`; shuttle +0.013 CI `[-0.002, +0.029]`; gate_a -0.001 CI `[-0.012, +0.008]`; person -0.009 CI `[-0.045, +0.021]`.

205 - Gate A served default is ****rejected****: proposal mean Delta F1 +0.0287, CI `[-0.0222, +0.0680]`; served mean Delta F1 -0.0328, CI `[-0.0799, +0.0052]`, `regresses=True`.

206 - R2/cap/inactivity remains the strong quality result: served candidate F1 `0.472` vs baseline-off `0.237`, Delta F1 +0.236, CI `[+0.161, +0.313]`; tolF1 `0.364` vs `0.138`, Delta tolF1 +0.225, CI `[+0.165, +0.283]`; merge rate `0.135` vs `0.638`. Net over-seg guard passes, strict per-video split-count guard still blocks blanket promotion language.

207

208 A9 Gen-0 shadow-training record:

209

210 ```powershell

211 python -m training.gen0.harness --manifest output\human_lovo_manifest.json --floor

eval_baselines\heuristic_lovo_n15.json --out artifacts\rally_tal_gen0 --version

0.1.1-n15-shadow.20260704

212 ```

213

214 Result: current Gen-0 LogReg/TAL harness produced local bundle `artifacts\rally\_tal\_gen0\0.1.1-n15-shadow.20260704` with learned LOVO mean F1 `0.551` versus heuristic floor `0.4456`, so `floor\_passed=True`. The gate remains ****non-serving****: paired sign test `9/15`, `p = 0.3036`, `ship=False`; blockers are uncalibrated ship-gate target #172 and WASB checkpoint provenance C3. TemporalMaxer itself is not implemented in this branch; the evidence supports either closing A9 as a Gen-0 shadow baseline or opening a follow-up M0.4 scorer-swap PR.

215

216 ## 9. Open questions

217

218 1. ****Owner:**** Is the first alpha allowed to use manual/offline corpus promotion, or should P10/P11 be reopened before any tester wave?

219 2. ****Owner/platform:**** Which hosted compute target is the next stand-up: Cloud Run GPU,

GCE L4 VM, or another approved GPU host?

220 3. ****Owner/legal:**** What is the acceptable alpha wording while WASB C3 remains unresolved?

221 4. ****Engineering:**** Should the portable corpus manifest live as a generated artifact, a checked-in redacted manifest, or both?

222 5. ****Engineering:**** Should A9 close on the current Gen-0 non-serving bundle, or should the actual TemporalMaxer scorer swap be implemented before marking A9 done?

223

224 ## 10. Definition of done

225

226 - [] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.

227 - [] Alpha service is gated behind the chosen auth posture and has a documented disable/rollback path.

228 - [x] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale local absolute paths.

229 - [x] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.

230 - [x] R2/tolF1 gate is promoted or rejected with a recorded ablation result.

231 - [x] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the 15-video corpus.

232 - [] Ship-gate target and allowed alpha claims are recorded.

233 - [] WASB C3 status is recorded and reflected in alpha messaging.

234 - [] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-plus.

235 - [] Owner records GO/NO-GO for the first alpha wave.

236

237 ## 11. References

238

239 Internal docs:

240

241 - `docs/NEXT\_STEPS.md`

242 - `docs/ROADMAP.md`

243 - `docs/data\_pipeline/GOLDEN\_VIDEOS.md`

244 - `docs/R2\_EVAL\_METRIC\_DESIGN.md`

245 - `docs/RALLY\_DETECTION\_FIXES\_PLAN.md`

246 - `docs/COMPUTE\_DECOUPLED\_SERVING/README.md`

247 - sibling `khehsutra/NEXT\_STEPS.md`

248 - sibling `khehsutra/docs/STUDIO\_MEDIA\_LIFECYCLE.md`

249 - sibling `khehsutra/docs/PER\_RALLY\_SELECTION.md`

250 - sibling `sports-data-collector/data/golden\_source\_videos.json`

251

252 Local verification commands used for the 2026-07-03 snapshot:

253

254 ```powershell

255 python -m backend.eval.fusion_compare --features-dir output

256 python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>

257 python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor

eval_baselines\heuristic_lovo_n15.json --version analysis-n15 --out <temp-out>

258 python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir

output --no-report

259 python -m backend.eval.ablation --features-dir output --granularity group

260 python -m backend.eval.rally_seg_eval --manifest output\human_lovo_manifest.json

--features-dir output --preset served --max-window-duration 0 --inactivity-end 0 --vs served

--vs-max-window-duration 6 --vs-inactivity-end 1.5 --json-out output\A6_r2_ablation.json

261 ```

262

263 ---

264

265 ### Changelog

266

267 - 2026-07-03 - Created from a synced five-repo review of `NEXT\_STEPS`, `ROADMAP`, golden-corpus docs, local eval artifacts, and the current 15-video corpus state.

268 - 2026-07-04 - Recorded A7 full-fusion/audio regeneration, A8 n15 fusion/default decisions, and A9 Gen-0 n15 shadow-training gate verdict.

269 - 2026-07-04 - Marked A3/A7/A8/A9 complete after collector #62 and indexer #458

review/CI; clarified that the two Boxhill eval inputs are run-log-only 1080p hashes and that F-drive 4K originals are provenance-only.

270

6. user (2026-07-04T11:48:51.190Z)

Found 12 files

C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\COMPUTE_DECOUPLED_SERVING\runlogs\DEPLOY_REPORT_cloud-serving_a1-cuj_2026-07-04.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\archives\past_projects\AUDIT_REPORT_khealsutra-guru_2026-07-02-COMPLETED-2026-07-02.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\CODE_MAP.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\HOSTING_PLAN.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\I18N_PLAN.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\DECISION_SNAPSHOT_REGRESSION.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\UI_PROPOSAL.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\PACKAGING_PLAN.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\COMPUTE_DECOUPLED_SERVING\README.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md

7. user (2026-07-04T11:48:52.599Z)

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C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\archives\past_projects\AUDIT_REPORT_khealsutra-guru_2026-07-02-COMPLETED-2026-07-02.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\HOSTING_PLAN.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\DESKTOP_APP_PLAN.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\data_pipeline\VIDEO_RECORDING_GUIDELINES.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\RALLY_DETECTION_CODE_REVIEW.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\DECISION_SNAPSHOT_REGRESSION.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\RALLY_QUALITY_RESEARCH.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\R1_MILESTONE_LAUNCH_REPORT.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\RALLY_DETECTION_QUALITY_REPORT.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\PROMINENT_COURT_DETECTION.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\COMMERCIALIZATION.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\PERSONALIZATION_PLAN.md
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\ANALYZERS_AND_RECONCILERS.md
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C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\docs\archives\MONETIZATION_AUDIT.md

8. user (2026-07-04T11:48:53.446Z)

No matches found

9. user (2026-07-04T11:49:04.630Z)

```
1      # A1 alpha serving deploy ? hosted CUJ run report (cloud-serving)
2
3      **Date:** 2026-07-04 . **Region:** asia-southeast1 . **Project:** gen-lang-
client-0306026826
4      **Scope:** A1 (ALPHA_LAUNCH_READINESS $7) ? stand up the alpha serving endpoint and
verify the
5      hosted `process ? jobs ? compile ? download` CUJ against the WASB-only owned detector.
6
7      > Companion to `DEPLOY_REPORT_cloud-serving_2026-07-04.md` (PR #457, the GPU-batching
runlog). That
8      > report proved the L4 worker + measured the batching win via the M6 dispatch **shim**;  
THIS report
9      > proves the deployed **control-plane service** end-to-end. They do not overlap.
10
11     ## 1. What was deployed
12
13     | Resource | Value |
14     |---|---|
15     | L4 worker **Job** | `rally-worker` ? nvidia-l4, 8 vCPU / 32Gi, `--no-gpu-zonal-
redundancy`, `--max-retries 0 --task-timeout 3600`, GCS-FUSE mount `khesutra-rally-
serving` ? `gcs`, `WASB_WEIGHTS=/gcs/models/wasb_badminton_best.pth.tar` |
16     | Control-plane **Service** | `rally-control-plane` ? rev 00002, `--no-cpu-throttling`,
`min/maxScale 1/3`, cpu 2 / mem 4Gi, port 8080, **private (IAM-gated, no allUsers)** |
17     | Service image | `asia-southeast1-docker.pkg.dev/?/rally/rally-control-plane:latest` (=
`rally-worker:latest` + `google-cloud-run`; see $5 bug 1) |
18     | Service URL | `https://rally-control-plane-2bosgeukpa-as.a.run.app` |
19     | Config (via base64 startup wrapper ? `RALLY_APP_HOME=/tmp/apphome/config.json`) |
`deployment.profile=cloud-serving`, `compute_target=cloud_run`, `indexing.skip_ai_handoff=true`,
`storage.backend=gcs`, `output_root=gs://khesutra-rally-serving`, `input_backend=gcs`,
`input_root=?/videos` |
20     | Env | `DEPLOYMENT_PROFILE=cloud-serving`, `CLOUD_RUN_REGION=asia-southeast1`,
`GOOGLE_CLOUD_PROJECT=?`, `RALLY_ALLOWED_HOSTS=*` (CLOUD_RUN_JOB is reserved ? code default
`rally-worker` is correct) |
21
22     **Boot-crash trap avoided:** the `cloud-serving` preset forces
`indexing.skip_ai_handoff=False`
23     (Gemini ON), which would trip the `AI_HANDOFF_KEY_MISSING` startup check (no Gemini key)
and
24     crash-loop. Verified `loader.py:194` deep-merges `defaults < preset < config.json`, so
the
25     explicit `skip_ai_handoff=true` wins. Also pinned `storage.backend=gcs` explicitly to
satisfy the
26     `STORAGE_INCOHERENT` check. Boot log confirmed: `deployment profile ACTIVE: cloud-
serving
27     (compute_target=cloud_run)` + `Uvicorn running on http://0.0.0.0:8080`.
28
29     ## 2. Health / capabilities (authenticated via `gcloud auth print-identity-token`)
30
31     - `GET /api/health` ? `200
{"status":"ok","sport":"badminton","default_segmenter":"yolo_hybrid}`
32     - `GET /api/capabilities` ? `segmenters` incl. `wasb_hybrid`;
`deployment.cloud_available=true`;
33     `gemini_key_present=false` (no crash ? skip_ai_handoff honored); `gcs` backend usable.
34
35     ## 3. gs:// bucket-path CUJ ? PROVEN end-to-end
36
37     1. `POST /api/process {"video_path":"gs://khesutra-rally-
serving/videos/MahadevpuraSingles.MP4","segmenter":"wasb_hybrid","compute":"cloud"}` ? `202
{job_id: d16654d1?}`
38     2. Poll `GET /api/jobs/{id}`: `validating ? dispatching (cloud_run) ? done`.
39     - CP pulled + validated the video (format/resolution/pts/scene/blur/relevance all
```

```

pass).
40     - Dispatched L4 Job execution `rally-worker-k49kg`; worker log: `WASB in FULLY-LOCAL
mode
41     (skip_ai_handoff=True) ? no gemini handoff, no API key needed`, `torch 1.11.0+cu113
cuda True`.
42     - **Result: `success` ? "Successfully indexed 48 play segments (cloud_run /
'wasb_hybrid')***;
43     outputs at `gs://khelsutra-rally-serving/outputs/915473?/\`.
44     - Wall: created 02:12:42 ? finished 02:26:27 (**~13m45s**, incl. L4 cold image pull +
inference).
45     3. `POST /api/compile {video_id, segment_ids:[1..48],
output_filename:"mahadevpura_wasb_highlights.mp4"}` ? `202 {job_id: d947a9e6?}` ? poll ?
**`success`**. In-process ffmpeg stitch (trim 48 clips w/ padding ? concat ? watermark) on the
CP; **`Dynamic highlight reel saved ? (166.0 MB)`** ; wall ~3m26s (02:27:47?02:31:13, incl. re-
fetch of the gs:// source).
46     4. `GET /api/highlights/{video_id}/mahadevpura_wasb_highlights.mp4` ? the reel is served
(correct `206`/`video/mp4`/etag/size headers), and the **full 174,038,655-byte reel was
retrieved via <32 MiB Range requests and fprobes as a VALID video: H.264 1920x1080 @ 29.97fps +
AAC, duration 244.3s** (48 rallies). ? **A single full `GET` returns edge-500** ? Cloud Run's
~32 MiB response cap vs. the `FileResponse` fixed `Content-Length` (see $5 bug 3).
47
48     ## 4. Notes / observations
49
50     - **In-process compile serializes behind process jobs** (drain
`ThreadPoolExecutor(max_workers=1)`):
51     a queued compile waits while a process job holds the worker in its bucket-poll. Fine
for the alpha
52     (low concurrency); worth noting for load.
53     - **WASB batch tuning is NOT forwarded on the dispatch path.** `maybe_dispatch_remote`
forwards only
54     `skip_ai_handoff` (orchestration.py:274-283); the L4 job runs its baseline batch (not
the preset's
55     batch64/prefetch4), so the measured #457 ?30.5% win is not applied via the deployed
CP. Minor
56     (correctness unaffected); follow-up = forward `indexing.wasb.` overrides like
`skip_ai_handoff`.
57     - `player_pool` / `max_frames` are dropped on the cloud path (already noted at
orchestration.py:273).
58
59     ## 5. Bugs surfaced + handled
60
61     1. **`google-cloud-run` missing from the image** (BLOCKER). First real CP deploy failed
every dispatch
62     with **`google-cloud-run is required for CloudRunCompute's real client`** ? the image
installed only
63     `[storage-gcs]`; the reused-as-control-plane role needs `google.cloud.run_v2`.
**Unblocked** with a
64     derived image (`rally-worker` + `pip install google-cloud-run`). **Durable fix: PR
#462** (adds a
65     `cloud-run` extra + `[storage-gcs,cloud-run]` in the Dockerfile + test guard; 19
tests pass).
66     2. **Browser-upload ? gs:// staging gap.** Uploads land on the CP's local FS
(`/api/upload/complete` ?
67     `/tmp/apphome/output/uploads/?`), and `maybe_dispatch_remote` hard-rejects local
input under
68     cloud-serving (orchestration.py:250-255). Upload endpoints verified working; the gap
is staging.
69     **Filed: issue #461** (fix = stage local?gs://input_root before dispatch).
70     3. **`/api/highlights` download 500s for reels >32 MiB.** `FileResponse` sets a fixed
`Content-Length`;
71     Cloud Run caps a single buffered HTTP/1 response at ~32 MiB (empirically: 29.6 MB
Range ? 206,
72     32.4 MB ? 500). App logs 200; Google Frontend returns 500. Every real reel exceeds 32
MiB.
73     **Filed: issue #463** (fix = serve reels via a gs:// signed-URL redirect; also fixes

```

```

the ephemeral
74     `/tmp` reel location). Workaround verified: ranged download reassembles a byte-exact,
valid reel.
75
76     ## 6. M9 edge auth ? ENABLED + gate verified (DNS routing remains, owner-handoff)
77
78     Cloudflare Access edge-auth is **ON** (rev `rally-control-plane-00004-tvr`) and
enforcing:
79     - Image rebuilt to add `PyJWT[crypto]` (the `auth` extra) + the `google-cloud-run`
client.
80     - Config: `security.edge_auth_enabled=true`,
`cf_team_domain=https://kheIsutra.cloudflareaccess.com`,
81     `cf_access_aud=c980da67?082b` (the existing "KheIsutra Studio (alpha)" CF Access app
already lists
82     `api.kheIsutra.guru` in `self_hosted_domains` ? no new app needed; same 4-email
allowlist policy),
83     `allowed_video_root=/tmp/apphome/output`.
84     - **Bug found + fixed to make M9 compatible with cloud-serving:** enabling edge-auth
requires
85     `allowed_video_root`, and its jail (`validate_input_path`?`resolve_under_root`)
`realpath()`-mangled
86     and **rejected gs:// inputs** ? so edge-auth would have 400'd the proven gs:// CUJ.
**PR #465** makes
87     the local-path jail skip storage URIs. Overlaid into the deployed image (build-time
asserted);
88     runtime-verified (Phase 1): gs:// ? 202 accepted; a local out-of-root path ? 400 (jail
still enforced).
89     - **Verified (Phase 2):** `/api/health` (exempt) ? 200; `/api/process` without a CF JWT
? **401**
90     ("missing Cf-Access-Jwt-Assertion header"). Service boots (no crash-loop from the
exposure checks).
91
92     **Remaining (owner, CF/GCP dashboard ? outward-facing production DNS):**
`api.kheIsutra.guru` is still a
93     stale CNAME to a dead CF tunnel. To make the alpha reachable-with-auth by allowlisted
testers: (a) add
94     `allUsers`?`roles/run.invoker` on the Cloud Run service (so CF's proxied requests reach
the app ? the
95     app still requires the CF JWT, so direct run.app access stays 401); (b) route
`api.kheIsutra.guru` to the
96     service (Cloud Run domain-mapping in asia-southeast1, or a CF origin-rule Host
override); (c) re-point the
97     CF DNS record (proxied) off the dead tunnel. Then the authed CUJ works end-to-end
through the CF login.
98     Full end-to-end authed CUJ verification needs (a)+(b) (a CF-issued JWT only arrives via
the CF edge).
99
100    Rollback (make it owner-usable again without CF): redeploy with `EDGE_AUTH` unset (edge-
auth off) via
101    `scratchpad/gen_service_yaml.py` ? `gcloud run services replace`.
102
103    ## 7. Cost ledger + teardown
104
105    - One L4 execution (~14 min, 8 vCPU + 1xL4) for the CUJ. CP service holds `min-instances
1` (small
106    always-on CPU cost). Owner directive: **keep running** (closed alpha).
107    - Teardown (if needed): `gcloud run services delete rally-control-plane`, `gcloud run
jobs delete
108    rally-worker`, both `--region asia-southeast1`; bucket/secret/repo retained.
109

```

10. user (2026-07-04T11:49:06.582Z)

Found 4 files

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\locales\en\common.json

```
C:\Users\avidu\Projects\khehsutra-guru\khehsutra\site\public\locales\en\common.json
C:\Users\avidu\Projects\khehsutra-guru\khehsutra\site\public\privacy.html
C:\Users\avidu\Projects\khehsutra-guru\khehsutra\site\scripts\provision-vm.sh
```

11. user (2026-07-04T11:49:12.854Z)

```
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-
staging\deploy\gcp\nightly_regression\register_scheduler.sh:75:           gcloud beta run
jobs executions logs read <execution> --region=$REGION # launcher logs
C:\Users\avidu\Projects\khehsutra-guru\wt-upload-
staging\deploy\gcp\nightly_regression\README.md:111:- **Launcher logs**: `gcloud run jobs
executions list --job=nightly-regression-launcher --region=asia-south1` then `? executions logs
read <exec> --region=asia-south1`.
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\deploy\gcp\nightly_regression\register_scheduler.sh:75:           gcloud beta run
jobs executions logs read <execution> --region=$REGION # launcher logs
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\deploy\gcp\nightly_regression\README.md:111:- **Launcher logs**: `gcloud run jobs
executions list --job=nightly-regression-launcher --region=asia-south1` then `? executions logs
read <exec> --region=asia-south1`.
C:\Users\avidu\Projects\khehsutra-
guru\wt-465fix\deploy\gcp\nightly_regression\register_scheduler.sh:75:           gcloud beta
run jobs executions logs read <execution> --region=$REGION # launcher logs
C:\Users\avidu\Projects\khehsutra-guru\wt-465fix\deploy\gcp\nightly_regression\README.md:111:-
**Launcher logs**: `gcloud run jobs executions list --job=nightly-regression-launcher
--region=asia-south1` then `? executions logs read <exec> --region=asia-south1`.
C:\Users\avidu\Projects\khehsutra-guru\wt-signed-
download\deploy\gcp\nightly_regression\register_scheduler.sh:75:           gcloud beta run
jobs executions logs read <execution> --region=$REGION # launcher logs
C:\Users\avidu\Projects\khehsutra-guru\wt-signed-
download\deploy\gcp\nightly_regression\README.md:111:- **Launcher logs**: `gcloud run jobs
executions list --job=nightly-regression-launcher --region=asia-south1` then `? executions logs
read <exec> --region=asia-south1`.
C:\Users\avidu\Projects\khehsutra-guru\wt-dockerfile-
auth\deploy\gcp\nightly_regression\register_scheduler.sh:75:           gcloud beta run jobs
executions logs read <execution> --region=$REGION # launcher logs
C:\Users\avidu\Projects\khehsutra-guru\wt-dockerfile-
auth\deploy\gcp\nightly_regression\README.md:111:- **Launcher logs**: `gcloud run jobs
executions list --job=nightly-regression-launcher --region=asia-south1` then `? executions logs
read <exec> --region=asia-south1`.
C:\Users\avidu\Projects\khehsutra-guru\wt-drain-
concurrency\deploy\gcp\nightly_regression\register_scheduler.sh:75:           gcloud beta
run jobs executions logs read <execution> --region=$REGION # launcher logs
C:\Users\avidu\Projects\khehsutra-guru\wt-drain-
concurrency\deploy\gcp\nightly_regression\README.md:111:- **Launcher logs**: `gcloud run jobs
executions list --job=nightly-regression-launcher --region=asia-south1` then `? executions logs
read <exec> --region=asia-south1`.
C:\Users\avidu\Projects\khehsutra-guru\wt-input-
jail\deploy\gcp\nightly_regression\register_scheduler.sh:75:           gcloud beta run jobs
executions logs read <execution> --region=$REGION # launcher logs
C:\Users\avidu\Projects\khehsutra-guru\wt-input-
jail\deploy\gcp\nightly_regression\README.md:111:- **Launcher logs**: `gcloud run jobs
executions list --job=nightly-regression-launcher --region=asia-south1` then `? executions logs
read <exec> --region=asia-south1`.
C:\Users\avidu\Projects\khehsutra-guru\wt-cloudrun-
dispatch\deploy\gcp\nightly_regression\register_scheduler.sh:75:           gcloud beta run
jobs executions logs read <execution> --region=$REGION # launcher logs
C:\Users\avidu\Projects\khehsutra-guru\wt-cloudrun-
dispatch\deploy\gcp\nightly_regression\README.md:111:- **Launcher logs**: `gcloud run jobs
executions list --job=nightly-regression-launcher --region=asia-south1` then `? executions logs
read <exec> --region=asia-south1`.
```

12. assistant (2026-07-04T11:49:25.769Z)

Now let me pull the GitHub issues (#471, #472) read-only via gh, and look for Cloudflare Access

tester-management and teardown docs.

13. user (2026-07-04T11:49:28.267Z)

```
{"body":"## Symptom\n\nThe A1 control-plane service (rev 00004) runs with `min-instances 1` but  
**maxScale 3**. All server state ? the jobs queue, the videos/segments tables (SQLite), and  
scratch files ? lives in the instance's ephemeral `/tmp`. With more than one instance:\n\n-  
`/api/process` returns 202 from instance A; the client's `/api/jobs/{id}` poll can land on  
instance B, which has never heard of the job ? 404 / empty status while the job is actually  
running.\n- A restart or scale-down loses the queue outright (partially mitigated:  
`recover_stale_jobs()` re-queues interrupted compiles, but only on the instance that restarts,  
and completed-job history is gone).\n\n#463 (reel mirror to gs://) and #461 (upload staging to  
gs://) fix the **artifacts** outliving an instance, but not job/queue **state** visibility  
across instances.\n\n## Options\n\n1. **Alpha-cheap:** pin `maxScale=1` in the service manifest  
+ document the single-instance constraint (matches the current 4-email alpha traffic; `min-  
instances 1` already implies a warm single box).\n2. **Durable:** move job/video state to a  
shared store (Cloud SQL / Firestore / a GCS-backed job store). Pairs with #448 (durable  
background-job UX) and #326 (request tracing).\n\nUntil one lands, treat any multi-instance  
scale-out of the control plane as broken for the async-job CUJ.\n\nFound during the A1 alpha  
deploy (runlog #464).","number":471,"state":"OPEN","title":"cloud-serving: control-plane state  
(SQLite jobs/videos DB) is per-instance ephemeral ? maxScale\u003e1 splits the queue and  
/api/jobs polls"}
```

14. user (2026-07-04T11:49:29.795Z)

```
{"body":"## Watch-item from the A1 deploy (2026-07-04)\n\n`api.khelsutra.guru` fronts the Cloud  
Run control plane via a **domain mapping** (CNAME `ghs.googlehosted.com`, **PROXIED** through  
Cloudflare) + a Google-managed certificate. Initial issuance succeeded, but Google-managed certs  
renew ~every 90 days and the renewal validation can fail when the CNAME is proxied (Google's  
challenge sees Cloudflare's edge, not the expected record).\n\n## Action\n\n- **Before  
~2026-09-30:** check the domain-mapping cert status (`gcloud beta run domain-mappings describe  
--domain api.khelsutra.guru --region asia-southeast1`) and confirm renewal isn't stalled.\n- If  
it stalls: either temporarily grey-cloud (DNS-only) the CNAME so the renewal validation passes,  
or switch to a **Cloudflare Origin CA cert** on the mapping so renewal never depends on what  
Google can see through the proxy.\n\nContext: session notes from the A1 alpha deploy (runlog  
#464). CF Access (M9) is unaffected either way ? this is only about the Google-side TLS cert on  
the mapping.","number":472,"state":"OPEN","title":"ops: api.khelsutra.guru Google-managed cert  
auto-renew (~90d) can stall behind the proxied CF CNAME ? verify before ~2026-09-30"}
```

15. user (2026-07-04T11:49:29.903Z)

5:acceptable · alpha backend = owner's GPU box behind a tunnel · 10?25 alpha testers · registrar
= GoDaddy.
14:**Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU box)
at
19:Tunnel + Access combo (IAP requires a load balancer, ~\$18/mo standing cost), and the alpha
needs
21:a custom domain, email-allowlist auth (free ?50 users ? fits 10?25 testers), and static
hosting.
27:tester browser
30:Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
41:- **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
42: `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-user
scoping.
47:- **Costs:** Pages, Tunnel, Access, DNS = \$0. Only the Gemini per-run cost (~\$0.05/match) and
67: later); Access allowlist retires.
95:| Access | Scope | Used for |
97:| **Cloudflare API token** (you create: dash ? My Profile ? API Tokens) | Zone:DNS:Edit,
Account:Pages:Edit, Account:Access:Apps:Edit, Account:Tunnel:Edit ? scoped to this zone | DNS
records, Pages deploys (wrangler), Access policies, tunnel config |
103:API/tunnel/DNS changes via the scoped token. Secrets (Gemini key, Access policy) never live
in the
108:Access allowlist ON for both hostnames ? CORS narrowed to `https://app.khelsutra.guru` ?
rate

110:allowlist ? schedule upload retention for `security.upload\_dir` with `python -m tools.upload\_retention`

16. assistant (2026-07-04T11:49:43.981Z)

Issues confirmed. Now the teardown conventions, memory rules, CF Access management details, and the cloudbuild config.

17. user (2026-07-04T11:49:44.933Z)

```
1      # Cloud Build config for the Cloud Run GPU worker image (COMPUTE_DECOUPLED_SERVING M7).
2      #
3      # `gcloud builds submit --tag ? -f ?` does NOT work ? `gcloud builds submit` has no `-f`
flag, so a
4      # non-root Dockerfile needs a build CONFIG like this one. Build with:
5      #
6      #   gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project <PROJECT> .
7      #
8      # Override the region/repo/image with --substitutions=_REGION=?,_REPO=?,_IMAGE=? if
needed.
9      # PROJECT_ID is provided by Cloud Build automatically.
10     substitutions:
11         _REGION: asia-southeast1    # L4 GPU is region-limited (asia-southeast1 / europe-west1
/ europe-west4 / us-central1 / us-east4)
12         _REPO: rally
13         _IMAGE: rally-worker
14     steps:
15         - name: 'gcr.io/cloud-builders/docker'
16           args:
17             - build
18             - -t
19             - '${_REGION}-docker.pkg.dev/${PROJECT_ID}/${_REPO}/${_IMAGE}:latest'
20             - -f
21             - deploy/cloudrun/Dockerfile
22             - .
23     images:
24         - '${_REGION}-docker.pkg.dev/${PROJECT_ID}/${_REPO}/${_IMAGE}:latest'
25     options:
26         machineType: E2_HIGHCPU_8    # the CUDA + micromamba image is heavy; a bigger builder
keeps it ~8?10 min
27         timeout: 3600s
28
```

18. user (2026-07-04T11:49:46.460Z)

```
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\deploy\gcp\create_gpu_vm.sh:17:#      RALLY_VM_NAME=<name> RALLY_GCP_ZONE=<zone-it-
printed> bash deploy/gcp/teardown.sh delete
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-
indexer\deploy\gcp\create_gpu_vm.sh:52:  echo "    DELETE: RALLY_VM_NAME=$NAME
RALLY_GCP_ZONE=$Z bash deploy/gcp/teardown.sh delete"
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\deploy\gcp\teardown.sh:9:#
bash deploy/gcp/teardown.sh          # DELETE the VM (default; $0 ? removes VM + boot disk)
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\deploy\gcp\teardown.sh:10:#
bash deploy/gcp/teardown.sh delete    # delete the VM (no disk cost; full re-port to resume)
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\deploy\gcp\teardown.sh:11:#
bash deploy/gcp/teardown.sh stop      # STOP only ? keeps the disk ? STILL BILLS (~$/mo!); fast
resume
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\deploy\gcp\teardown.sh:12:#
bash deploy/gcp/teardown.sh all       # delete VM + SA + budget (keeps the bucket/data)
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\deploy\gcp\teardown.sh:14:#
Cross-region VM: set the zone, e.g.  RALLY_GCP_ZONE=asia-southeast1-a bash
deploy/gcp/teardown.sh
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:18:> for
```


its disk), and keep the **\$100 budget alert** live. `teardown.sh` switches everything off.

```
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:38:|
**GPU VM** | **NOT created** ? blocked on `GPUS_ALL_REGIONS` quota (see $2) | **~$0.35/hr (T4)
running, ~$0.01/hr stopped** | `teardown.sh` (stop or delete) |
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:41:Run
`bash deploy/gcp/teardown.sh` (see $6) to stop/delete the billable bits between sessions.
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:248:>
$0. So `teardown.sh` **defaults to `delete`** (owner cost rule, 2026-06-20). The boot disk has
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:252:bash
deploy/gcp/teardown.sh          # DELETE the VM (default; $0 ? removes VM + boot disk)
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:253:bash
deploy/gcp/teardown.sh delete   # same (explicit)
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:254:bash
deploy/gcp/teardown.sh stop     # STOP only ? keeps the disk ? STILL BILLS (~$/mo!); fast resume
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:255:bash
deploy/gcp/teardown.sh all      # delete VM + SA + budget (keeps the bucket/data)
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer\deploy\gcp\README.md:256:#
cross-region VM: set the zone (teardown defaults to asia-south1-a)
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-
indexer\deploy\gcp\README.md:257:RALLY_GCP_ZONE=asia-southeast1-a bash deploy/gcp/teardown.sh
delete
C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-
indexer\deploy\gcp\nightly_regression\run_on_vm.sh:35:# ALWAYS delete the VM on exit (success OR
failure) ? never leave a billing GPU VM behind.
```

19. user (2026-07-04T11:49:48.474Z)

```
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\cloudrun-gpu-
serving-gotchas.md:13:`gs://kheIsutra-rally-corpus` (asia-south1). Pairs with [[compute-stack-
gcp-only]] [[gcp-vm-always-delete]].
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\compute-stack-gcp-
only.md:19:- The shared DO droplet `claude-do-rally-test` (other session's testbed) + retiring
the DO API token = owner-coordinated, separate from this. See [[gcp-vm-always-delete]] [[gcp-
ops-agent-enable]] [[monetization-plan]].
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\compute-stack-gcp-
only.md:21:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\gcp-vm-always-
delete.md:2:name: gcp-vm-always-delete
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\kheIsutra-site-
droplet-parity.md:54:[[compute-stack-gcp-only]] (was the ~$24/mo non-compute DO testbed) +
[[gcp-vm-always-delete]].
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\MEMORY.md:33:-
[GCP VM: always DELETE, never just stop](gcp-vm-always-delete.md) ? owner cost rule
(2026-06-20): a STOPPED VM still bills its disk (~$34/mo for 200GB pd-ssd); **DELETE** ? $0
(boot disk autoDelete=true). `teardown.sh` now defaults to `delete`; audit `gcloud compute disks
list` for orphans. DO droplet `claude-do-rally-test` (~$24/mo) is a shared testbed (owner-
gated).
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\next-session-
packaging-app.md:18:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\next-session-
packaging-app.md:25:- Memory `[[packaging-app-plan]]` (full per-PR detail) . `[[gcp-vm-always-
delete]]` . `[[compute-stack-gcp-only]]` . `[[gotcha-shared-checkout-branch-collisions]]` .
`[[lesson-verify-engine-surface-before-scoping]]`.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\packaging-app-
plan.md:37:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\ui-driven-cloud-
serving-gap.md:65:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-kheIsutra-guru\memory\wasb-l4-perf-cpu-
feed-bound.md:63:[[feedback-tail-long-job-logs]] for how to watch the L4 runs. Related: [[gcp-
vm-always-delete]].
```

20. user (2026-07-04T11:49:49.309Z)

```
C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\web\README.md:1:# web/ ? KheIsutra Studio
(operator console SPA)
```

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\web\index.html:6: <title>KheIsutra Studio</title>

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\docs\ANNOTATE_AND_TRAIN.md:1:# Annotate your footage ? train your own model (KheIsutra Studio USP pages)

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\NEXT_STEPS.md:82:- **Single-box appliance (KheIsutra Studio)** ? *** Tracked design: `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` (DRAFT, owner-gated).**

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\web\package.json:6: "description": "KheIsutra Studio ? operator console SPA (drives the rally indexer engine).",

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:1:# Single-box appliance ? UI-powered rally indexer + reel generator (KheIsutra Studio)

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:16:[Omitted long matching line]

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\docs\STUDIO_MEDIA_LIFECYCLE.md:23:Turn five independent CLIs into ONE seamless KheIsutra Studio flow: **Add a video ? store it in a user?chosen medium

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\docs\README.md:23: generator (KheIsutra Studio): the `web/` SPA drives the full ingest ? index ? reel loop on one

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\web\e2e\README.md:3:A hermetic, single-origin **end-to-end proof + screencast** of KheIsutra Studio: the real `web/` SPA

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\GPU_VM_SERVING.md:1:# Move KheIsutra Studio onto an L4 GPU VM

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\GPU_VM_SERVING.md:58: the kheIsutra app ? its `AUD` tag** ; team domain is `https://<your-team>.cloudflareaccess.com`.

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\GPU_VM_SERVING.md:149:team domain, and hosts are the same values** the retired droplet deploy used (same tunnel + Access app;

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\GPU_VM_SERVING.md:240:DNS, the CF-Access app, and the dashboard ingress are unchanged ? only the connector host moved.

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\droplet\README.md:15:`api.kheIsutra.guru`**, team domain

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\droplet\README.md:54:# cf_access_aud is still a placeholder ? you fill it in step C after creating the Access app.

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\droplet\config.json.example:6: "cf_team_domain": "https://kheIsutra.cloudflareaccess.com",

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\cloudflared.config.example.yml:1:# cloudflared tunnel config ? KheIsutra Studio ENGINE (api.kheIsutra.guru ? FastAPI on the box).

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\DEPLOY_ALPHA.md:1:# Deploy KheIsutra Studio for alpha ? a shareable page, no SSH

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\DEPLOY_ALPHA.md:7:> **Still pending (owner, Cloudflare dashboard):** clean up the dangling `app.`/`api.kheIsutra.guru` CNAMEs and the CF-Access application ? they still resolve via Cloudflare to a tunnel with no origin.

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\DEPLOY_ALPHA.md:65: "cf_team_domain": "https://<your-team>.cloudflareaccess.com",

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\DEPLOY_ALPHA.md:66: "cf_access_aud": "<the CF Access application AUD>" // filled in \$4

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\DEPLOY_ALPHA.md:114:4. **Copy the application `AUD` tag** ? paste into the engine's `config.json` `security.cf\_access\_aud` (\$2.2), and set `cf\_team\_domain` to `https://<your-team>.cloudflareaccess.com`. Restart the engine.

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\DEPLOY_ALPHA.md:115:5. **CORS preflight ?** the browser sends an unauthenticated `OPTIONS` preflight to `api.` for the cross-origin POSTs. In the Access application's **Settings ? enable ?Bypass options requests to CORS preflight** (or add an explicit OPTIONS bypass), else the preflight is redirected to the login page and every API call fails. The engine's CORS middleware (`allow\_credentials=false`, no cookies) then answers the preflight.

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\deploy\DEPLOY_ALPHA.md:156:`cloudflared tunnel cleanup kheIsutra-studio` + delete the tunnel; remove the Pages custom domain; disable the Access application; stop the engine service. The box's `output/` + `uploads/` retain originals ? sweep them per your retention policy before any wider exposure.

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\web\src\i18n\index.ts:1:// i18n scaffold (tracker B1) ? the shared localization foundation for KheIsutra Studio.

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\site\public\flyer-es.html:111:

C:\Users\avidu\Projects\kheIsutra-guru\kheIsutra\site\public\flyer-id.html:111:
C:\Users\avidu\Projects\kheelsutra-guru\kheelsutra\site\public\flyer-da.html:111:
C:\Users\avidu\Projects\kheelsutra-guru\kheelsutra\site\public\flyer.html:113:
```

## 21. user (2026-07-04T11:50:01.405Z)

```
95 The SPA ships as a **static Pages site**, git-connected like the marketing site (auto-
deploy on merge to `master`).
96
97 1. **Cloudflare dashboard ? Workers & Pages ? Create ? Pages ? Connect to Git** ? pick
the `kheelsutra` repo.
98 2. **Build settings:**
99 - **Root directory:** `web`
100 - **Build command:** `npm run build`
101 - **Build output directory:** `web/dist`
102 - **Environment variable:** `VITE_API_BASE = https://api.kheelsutra.guru/api` ? this
is what points the SPA at the engine tunnel (PR #64).
103 3. **Custom domain:** add `app.kheelsutra.guru` to the Pages project.
104
105 > Every push to `master` now rebuilds + redeploys the SPA. To preview a change first,
Pages builds a per-PR preview URL.
106
107 ---
108
109 ## 4. Cloudflare Access (gate BOTH hostnames)
110
111 1. **Zero Trust ? Access ? Applications ? Add ? Self-hosted.**
112 2. **Application domain:** add **both** `app.kheelsutra.guru` **and**
`api.kheelsutra.guru` to the *same* application (one app ? one shared session cookie, so a tester
signs in once and both the SPA and its API calls are authorized).
113 3. **Policy:** Action *Allow*, rule *Emails* ? your alpha testers' addresses (free ? 50
users ? fits 10?25 testers). Use one-time-PIN (email OTP) so testers need no Cloudflare account.
114 4. **Copy the application `AUD` tag** ? paste into the engine's `config.json`
`security.cf_access_aud` (§2.2), and set `cf_team_domain` to `https://<your-
team>.cloudflareaccess.com`. Restart the engine.
115 5. **CORS preflight ?:** the browser sends an unauthenticated `OPTIONS` preflight to
`api.` for the cross-origin POSTs. In the Access application's **Settings ? enable ?Bypass
options requests to CORS preflight?*** (or add an explicit OPTIONS bypass), else the preflight is
redirected to the login page and every API call fails. The engine's CORS middleware
(`allow_credentials=false`, no cookies) then answers the preflight.
116
117 ---
118
119 ## 5. The tunnel (engine ? Cloudflare)
120
121 ```bash
122 cloudflared tunnel login # browser auth to your zone
123 cloudflared tunnel create kheelsutra-studio # note the <TUNNEL_UUID>
124 # copy deploy/cloudflared.config.example.yml ? /etc/cloudflared/config.yml, fill
<TUNNEL_UUID>
125 cloudflared tunnel route dns kheelsutra-studio api.kheelsutra.guru
126 cloudflared tunnel run kheelsutra-studio # or install as a service:
`cloudflared service install`
127 ```
128 See `deploy/cloudflared.config.example.yml` for the ingress (only `api.kheelsutra.guru` ?
http://127.0.0.1:8000`).
129
130 ---
131
132 ## 6. Verify (the posture test + the CUJ)
```

```

133
134 **Exposure posture (must hold ? this is what makes it safe to share):**
135 - From OUTSIDE the box: `curl --max-time 5 http://<box-public-ip>:8000/api/health` ?
refused / timeout (the engine is `127.0.0.1`-bound; no inbound port). ? The box is not
directly reachable.
136 - `curl -i https://api.khelsutra.guru/api/health` **without** a CF session ? a **302 to
the Access login** (not a 200). ? The gate is in front.
137 - Open `https://app.khelsutra.guru` in a browser ? CF Access login ? after sign-in, the
SPA loads and `/api/health` returns 200 (same session cookie). ?
138
139 **The shareable CUJ (what a tester does):** open `app.khelsutra.guru` ? sign in ?
upload a video from their device (the SPA chunked-upload lands it in `upload_dir`, inside
the jail) ? **Process** ? watch progress ? pick rallies ? **Build** ? **download the reel**. All
from the page. *(Pasting a `gs://` URI does NOT work on this box ? see §7: the
`allowed_video_root` jail rejects every non-local URI. Browser upload is the alpha path.)*
140
141 ---
142
143 ## 7. Honest caveats / deferred
144
145 - **`upload_dir` must sit UNDER `allowed_video_root`.** If you enable browser upload, an
upload landing outside the jail is rejected by `/api/process` (a functional break, not a hole).
Keep `upload_dir` inside the jail (§2.2).
146 - **`gs://` / `s3://` / `gdrive://` URIs are REJECTED on the exposed box.** Setting
`allowed_video_root` (required to expose, §2.2) path-jails `/api/process` to *local* paths under
that root; the engine resolves a `gs://` URI to a local path that escapes the jail ? `400`
(pinned by engine `tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri`).
Cloud-URI ingest works **only** when `allowed_video_root` is unset (the local single-user
posture ? the opposite of this runbook). For alpha, the ingest path is **browser upload** (it
lands inside the jail). *(Scheme-exempting recognized buckets from the jail would be an engine
change, not a config flip.)*
147 - **You MUST set `edge_auth_enabled=true` to expose.** The boot posture (and the in-app
JWT verify) only engage then. Do not put the engine behind the tunnel with edge auth off.
148 - **Cross-origin reel download:** the reel streams from `api.` with `Content-
Disposition: attachment`, so it downloads even cross-origin ? but verify the filename in your
browser during the CUJ (cross-origin `` filename hints are ignored; the header is
the authority).
149 - **Upload size:** Cloudflare's free proxy caps request bodies at **100 MB** ? large
matches upload as ~95 MB chunks (the SPA chunked-upload path honours the engine's `max_part_mb`,
default **95** ; lower `security.max_part_mb` to shrink them). `gs://` bucket pre-staging is
not an option on this box (see the jail caveat above).
150 - **Box offline = engine offline.** The Pages SPA stays up and degrades gracefully ?
today that means the SetupWizard's offline branch (driven by its `/api/capabilities` fetch),
not a persistent "engine offline" banner; the SPA has no `/api/health` ping (unimplemented ?
corrected 2026-07-02). Processing resumes when the box is back. Beta removes this with a queue
(`HOSTING_PLAN.md` §3).
151 - **Gemini = paid + uploads footage to Google.** Keep the key rotated; use it only for
clips you're OK sending.
152
153 ---
154
155 ## 8. Teardown
156 `cloudflared tunnel cleanup khelsutra-studio` + delete the tunnel; remove the Pages
custom domain; disable the Access application; stop the engine service. The box's `output/` +
`uploads/` retain originals ? sweep them per your retention policy before any wider exposure.
157
158 ---
159 ### Changelog

```

## 22. user (2026-07-04T11:50:02.475Z)

Found 6 files

```

c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\MEMORY.md
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\session-handoff.md

```

```
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\ui-driven-cloud-
serving-gap.md
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\next-session-ux-
localization.md
```

### 23. user (2026-07-04T11:50:07.564Z)

```
1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for khelsutra-guru."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state - khelsutra-guru
11
12 **Updated:** 2026-07-04
13
14 ## Live Checkpoint
15
16 **2026-07-04 (SESSION END: A1 DONE ? ? e2e PASSED on the final image; edge-auth
restored; runlog+flip
17 MERGED #477; master `122f174`).** Post-#476 rebuild (`31475f27?`) ? job re-pinned + CP
rev 00006
18 (? `services replace` w/ unchanged manifest = NO-OP, used `services update --image` to
re-resolve
19 :latest) ? redeploy WIPED the ephemeral DB (the #471 bite: /api/highlights 404s at
main.py:1052 before
20 the bucket check) ? re-ran the FULL CUJ on the final image: gs:// process **48 rallies**
? . upload
21 staged+dispatched+done ? (#461 on both builds) . 2 process jobs drained concurrently ?
(#468) . compile
22 + gs:// mirror ? . **463 VERIFIED: 307 ? storage.googleapis.com v4 signed URL (signed
AS the compute
23 SA, keyless signBlob) ? full 174,046,228 bytes in 8.1s single-GET, ffprobe-valid
1080p29.97 244.29s** .
24 edge-auth ON restored (rev **00007**, same digest): health 200 / process-sans-JWT 401 /
25 api.khelsutra.guru ? CF-Access login 302 (right aud). Runlog + ALPHA_LAUNCH_READINESS $7
A1?DONE
26 merged as **477** on 11/11 CI (+ full local gates). All session worktrees/branches
cleaned (sibling
27 6e9d0958 worktrees untouched; prior session's stale non-git `wt-a1-runlog` dir removed).
Live stack KEPT
28 RUNNING per owner. ? NEXT per [[session-handoff]]: A13 launch review . khelsutra UX
backlog (#329 async
29 compile) . #471 durable state / #472 cert watch / generator `--revision-suffix`.
30
31 ---
32
33
34 **2026-07-04 (e2e MID-RUN: CUJ mostly GREEN; #463 signed-URL found BROKEN on real creds
? #475 filed ?
35 #476 FIXED+MERGED ? rebuilding).** e2e scoreboard: ? image rebuilt from merged master +
`rally-worker`
36 job re-pinned + CP **rev 00005** deployed via the committed generator (auth-off; base
image direct ?
37 proves #470, no derived overlay) . ? health/capabilities . ? gs:// CUJ **48 rallies**
(matches prior
38 run) . ? compile e2e.mp4 + **mirrored to gs:// (165.98 MiB)** . ? **461 VERIFIED**
720p upload
39 staged to `videos/<id>/?` (byte-exact) + dispatched + done (0 segments = correct for
```

```

synthetic; NB
40 1st attempt failed validation ? clip must be ?1280x720) . ? #468 live-proven (upload job
dispatched
41 WHILE compile stitched) . ??? **#463 redirect 500'd on real creds**: keyless signBlob
rode the
42 storage client's devstorage-scoped token (iamcredentials rejects; bare-except hid it;
fell back to
43 local serve ? 32MiB cap). Root-caused live (SA self-tokenCreator ?, iamcredentials API
?, scope ?)
44 ? **issue #475** ? **PR #476 MERGED** (`3c97117`: `_signblob_identity()` mints a fresh
cloud-platform
45 -scoped ADC cred; loud logging; 1659? 85.46% ruff/mypy clean; merged on green CI per
46 working-agreement-merging). ? NOW: rebuild image (bg) ? jobs update + services replace ?
47 re-verify #463 (307 + full >32MiB download) ? edge-auth ON restore ? api.khelsutra.guru
posture ?
48 runlog PR ? flip A1 DONE.
49
50 ---
51
52
53 **2026-07-04 (ALL 5 PRs MERGED on owner LGTM; e2e STARTED).** Owner posted written
logic-LGTMs on
54 #467/#468/#469/#470/#474 (~09:47Z) + authorized verify-and-merge in-session. Merged
SEQUENTIALLY, each
55 after a local **merge-result** verification in a temp worktree (full suite w/ coverage +
ruff + mypy;
56 per the pre-LGTM gate): baseline master 85.40% ?#467 (1637?, 85.38%, ?0.02pp = new
defensive except
57 branch, noise) ? #468 (1645?, **85.44%**) ? #469 (1652?, 85.39%) ? #470 (1652? + md-
links) ? #474
58 (1655?, 85.39%). No textual/semantic conflicts anywhere (the feared
orchestration.py/main.py overlaps
59 auto-merged clean + suite-proven). Master `195932f`; issues #461/#463/#466/#473 AUTO-
CLOSED; all 6
60 session worktrees + local branches removed (sibling session's 6e9d0958 worktrees left
alone).
61 **e2e (B) IN PROGRESS:** GCP pre-flight green (project/auth/job/service/bucket+test-
video); worker-image
62 rebuild from merged master SUBMITTED to Cloud Build (bg); service YAMLS generated from
the NEW committed
63 `deploy/cloudrun/gen_service_yaml.py` (auth-off + auth-on, config verified == handoff
spec). Next: build
64 green ? `gcloud run jobs update rally-worker --image :latest` (re-pin digest) ?
`services replace`
65 auth-off ? CUJ (#463 signed-URL >32MiB + #461 upload staging) ? auth-on restore ? runlog
? flip A1.
66
67 **2026-07-04 (PICKUP SESSION) ? #465 MERGED on owner LGTM; all 3 open review blockers
FIXED+PUSHED+REPLIED;
68 #470 opened; issues #471?473 filed.** Continuing session picked up the handoff: (1)
#465 merged (squash,
69 owner LGTM 08:55Z, 11/11 CI green ? master moved to the merge commit; #467/#468/#469
stayed MERGEABLE, no
70 conflicts). (2) **All three open blockers fixed** via a 3-agent workflow + adversarial
verify (each verifier
71 PROVED the new regression test fails at the unfixed PR head): **#468 ? 21ee141**
(claim_due_job retries a
72 lost CAS race; None now = truly empty; 8-claimer barrier test, 5/5 fail-without-fix,
15/15 pass-with),
73 **#467 ? c9bf37e** (staging gate classifies input_root by scheme via parse_uri, not
input_backend; reviewer's
74 gs://-root+default-backend repro now stages), **#469 ? ce450e5** (signed_url helper
ttl=None defers to
75 configured signed_url_ttl; GCS+S3+/api/highlights e2e TTL tests). Full suite + ruff
0.15.16 + mypy 2.1.0

```

76 green on every branch; replies posted on each PR. (3) \*\*PR #470\*\* (bake `auth` extra  
into the cloudrun base  
77 image ? kills the derived-PyJWT-image step of the e2e). (4) Rough edges filed: \*\*#471\*\*  
(per-instance SQLite  
78 splits CP state at maxScale>1), \*\*#472\*\* (api.khelsutra.guru Google-managed cert renewal  
watch-item, check  
79 before ~2026-09-30), \*\*#473\*\* (commit gen\_service\_yaml.py ? CP deploy not reproducible  
from clone). ? NEXT =  
80 LGTMs on #467/#468/#469/#470 ? squash-merge each (watch conflicts: #467/#469  
orchestration.py, #468/#469  
81 main.py) ? full e2e per [[session-handoff]] (B) (with #470 merged, SKIP the derived-  
image step ? deploy CP  
82 from the base image directly). Monitor loop was polling the PRs every 150s this session.  
83 \*\*Later same session:\*\* #470 went CI-red on first push  
(`test\_dockerfile\_has\_required\_directives` pins the  
84 exact extras string ? the #465 full-suite-before-push lesson repeated; owned it on the  
PR) ? guard updated,  
85 head c63d122, now \*\*11/11 green\*\*. Also built + opened \*\*PR #474\*\*  
(`gen\_service\_yaml.py` manifest generator  
86 + structural tests + README §3a canonical path; closes #473, encodes #471 option-1  
maxScale=1 default) ?  
87 1637? suite/ruff/mypy/md-links green. \*\*5 PRs now await owner LGTM: #467 #468 #469 #470  
#474\*\*, all  
88 CI-green + MERGEABLE. e2e prep note: with #470+#474 merged, the e2e needs NO scratchpad  
artifacts ? rebuild  
89 worker image, `gen\_service\_yaml.py --edge-auth off`, `services replace`, CUJ, flip edge-  
auth on, runlog.  
90  
91 ---  
92  
93  
94 \*\*2026-07-04 (HANDOFF) ? A1 serving LIVE behind CF Access; 5 fix PRs in review; NEXT  
SESSION picks up  
95 review-resolution ? merge ? e2e.\*\* [[session-handoff]] fully specifies the pickup.  
Merged: \*\*#462\*\*  
96 (cloud-run extra), \*\*#464\*\* (A1 runlog). OPEN + MERGEABLE + CI-green + all review  
comments ADDRESSED,  
97 awaiting owner LGTM: \*\*#465\*\* (input-jail input\_root allowlist), \*\*#467\*\* (upload?gs://  
staging, closes  
98 #461), \*\*#468\*\* (drain concurrency ? RE-DESIGNED after LGTM-withdrawal:  
`\_LOCAL\_COMPUTE\_LANE` serializes  
99 local/compile, only remote polls parallelize; closes #466; ? offered a \*park\*-vs-\*block\*  
follow-up to the  
100 reviewer), \*\*#469\*\* (32MiB download ? gs:// signed URL, closes #463; keyless signBlob  
VERIFIED). Do NOT  
101 merge without LGTM; run the WHOLE suite before any fix-push (the #465 CI-red lesson).  
After all 4 merge ?  
102 rebuild worker image from master + derived PyJWT layer ? deploy (edge-auth OFF) ? CUJ  
verifying the >32MiB  
103 signed-URL download + the upload staging ? restore edge-auth ON ? flip A1 DONE.  
Deployed: rev 00004,  
104 api.khelsutra.guru LIVE (CF Access, 4-email allowlist). See [[session-handoff]].  
105  
106 ---  
107  
108  
109 \*\*2026-07-04 (earlier) ? A1 CORE SERVING DEPLOYED + gs:// CUJ PROVEN end-to-end.\*\*  
Deployed the L4 job  
110 `rally-worker` + control-plane service `rally-control-plane` (PRIVATE/IAM, `--no-cpu-  
throttling` +  
111 min-instances 1) on Cloud Run asia-southeast1. Verified boot (no crash ? config.json's  
112 `skip\_ai\_handoff=true` beats the cloud-serving preset per loader.py:194;  
storage.backend=gcs avoids  
113 STORAGE\_INCOHERENT), health, capabilities. \*\*gs:// CUJ green:\*\* `/api/process` (gs://  
video, wasb\_hybrid,

```

114 compute=cloud) ? L4 WASB fully-local ? **48 rallies** ? `/api/compile` ? **valid 166 MiB
1080p30 reel,
115 244s** (ffprobe-confirmed; delivered to owner). 3 hosted-serving bugs surfaced+handled:
PR #462
116 (image lacked `google-cloud-run` dispatch client ? worked around with a derived CP
image, durable fix in
117 PR), **461** (browser-upload not staged to gs://), **463** (`/api/highlights` 500s >32
MiB Cloud Run
118 cap). Runlog **PR #464**. **M9 edge-auth ENABLED + verified** (rev 00004: image +
PyJWT/google-cloud-run + the
119 PR #465 URI-jail fix overlaid; `/api/process` sans CF JWT ? 401, `/api/health` ? 200).
Found+fixed **PR
120 #465** (allowed_video_root jail realpath-rejected gs://, which edge-auth would've
broken). **A1 NOT flipped
121 DONE** ? remaining: CF **DNS routing** for api.khelsutra.guru (owner dashboard: allUsers
+ domain-map/CF
122 origin-rule + re-point the dead-tunnel CNAME) + #461/#463. See [[session-handoff]].
123 Windows: run gcloud from PowerShell (MSYS mangles `/gcs`); quote comma-dict flags.
124
125 ---
126
127
128 **2026-07-04 (later) ? A1 deploy IN PROGRESS, context handoff. See [[session-handoff]]
for the exact next commands.** De-risked + half-executed: GCP resources recreated (serving
bucket + weights + test video + gemini secret + `rally` repo), worker image building, compile-
reliability fix MERGED (#460), config designed (WASB-only, gcs output_root, base64 in the
handoff). Clean state: 0 jobs / 0 services deployed, ~$0 idle. Next session: create L4 job ?
deploy CP service (`--no-cpu-throttling --min-instances 1`) ? health ? CUJ (bucket path first) ?
CF auth ? DEPLOY_REPORT ? flip A1 DONE. This session also merged #456/#457/#459/#460, reviewed
#451/#452/#458/#62, filed the SPA upload-guard bug.
129
130 ---
131
132
133 **2026-07-04 ? A1 infra proven + GPU optimization DONE + all cloud torn down.** First
real Cloud Run
134 L4 serving run of the owned WASB detector (fully-local, zero Gemini) worked end-to-end
via the M6
135 dispatch shim. **Key finding: the L4 is CPU-FEED-bound, not GPU-bound (GPU <1% on every
config).**
136 Measured on MahadevpuraSingles (13418 frames, L4/8vCPU): baseline 727.9s ?
**batch64+prefetch4 =
137 506.1s (?30.5%/1.44x, parity BYTE-IDENTICAL)**; torch cu118 arch bump = 0.8% noise
(discarded, +traj
138 numeric drift); batch128 = OOM (24GB ceiling). **Shipped: PR #457** (batch/prefetch
knobs + tests +
139 `DEPLOY_REPORT_cloud-serving_2026-07-04.md`) MERGED to master `c5fc27c` on 11/11 green
CI; earlier
140 **456** (loud AI-key startup check + unbuffered worker logs) also merged. Remaining ~5x
needs a
141 PARALLEL-FEED change (scoped in report §3: chunk video, 1 DataLoader worker/chunk; then
more-vCPU
142 helps). **ALL GCP resources deleted** (4 jobs, rally image repo, khelsutra-rally-serving
bucket,
143 gemini secret) ? $0-residual audit clean; kept owner's rally-corpus/-nightly. Detail
144 [[wasb-l4-perf-cpu-feed-bound]]. **A1 full serving deploy (control plane + CF-auth flip)
NOT done ?
145 deferred when owner pivoted to optimize+teardown; CF token at G:\My
Drive\cloudflare.token.txt is
146 VERIFIED-active for when it resumes.**
147
148 ---
149
150
151 **2026-07-03 (later) ? PR #452 (A2 golden-manifest portability) REVIEWED, awaiting

```



author fixes.\*\* Codex-authored `backend/eval/golden\_manifest.py` resolver + 8 consumers strict-wired + tests/docs. Gates reproduced green on head `5a91701` (16052/6 skip, cov 85.44%, ruff+mypy clean; the 5 served-Gate-A owner-box tests now RUN ? the A2 promise works). 4 findings posted (review 4627357796): \*\*fusion\_audio.py:38 left on raw json.load\*\* (missed consumer, breaks A7 audio augment on stale paths ? top finding, in review body since the file isn't in the diff); test guard `\_golden\_specs` swallows ManifestPathError ? silent skip-all with generic reason; basename-shadow hardening (corpus verified collision-free today); DEFAULT\_MANIFEST duplicated vs rally\_seg\_eval. Design note: \$9 Q4 producer-side fix stays open. Local checkout left ON the PR branch pending fix verification. \*\*A1 pre-flight (owner lifted spend gate, \$300):\*\* gcloud authed (project `gen-lang-client-0306026826`, 0 services/jobs deployed, `rally` registry + corpus buckets exist), Gemini key at `G:\My Drive\gemini.api\_key.txt` VERIFIED working (never print it; ? Secret Manager), CF token at `G:\My Drive\cloudflare.token.txt` \*\*VERIFIED active\*\* (zone khelsutra.guru readable, DNS read OK ? stale `api.`/\`app.` CNAMEs exist from June deploy; Access API OK ? existing Access app "KheLSutra Studio (alpha)" on app.khelsutra.guru to mirror for api.; write scopes prove at first edit). \*\*A1 pre-flight ALL GREEN ? A1 execution STARTED 2026-07-03.\*\* SPA ingest screen exists (khelsutra `App.ingest` + /api/process|jobs client); khelsutra #329 sync-compile 524 is in A1 scope.

152

153 \*\*2026-07-03 ? PR #451 (alpha readiness tracker) reviewed ? fixed ? verified ? OWNER-MERGED as `5eb8755` (squash); local PR branch deleted, all 5 repos synced.\*\* ? Next grab = \*\*A2 corpus portability\*\* (ungated; manifest confirmed stale: points at dead `C:/Users/avidu/Projects/Annotation Setup/...`, data now under `khelsutra-guru/` + WSL + F:) and the \*\*A6 ablation circular-import fix\*\*. A3's records half is doable post-A2 (newer-9 source MP4s located on F: ? GX/BXH/mahadevpura folders); its vault/upload half stays owner-gated. \*\*A7 is ALSO grabbable locally post-A2\*\* (verified 2026-07-03: RTX 2070 Super 8GB + torch 2.6.0+cu124 CUDA=True in the repo env; all 9 trajectories in `khelsutra-guru/Annotation Setup/Trajectories/`; `fusion\_golden` person pass is windows-only single-pass + `fusion\_audio` --augment` is CPU; newer-9 schema = windowing-only candidates, needs regeneration not upconvert; 2 sources GX010137/GX010141 still to pin on F:, Boxhill\_Doubles?BXH-file mapping to confirm). Codex-authored `docs/ALPHA\_LAUNCH\_READINESS.md` (cross-repo alpha tracker; \$7 = execution source of truth) + index edits. Claude review posted 8 verified findings inline (table-breaking raw pipe in A12, ambiguous cross-repo "D13", missing CDS-M11 cross-link, spurious A4?A6 dep, analysis-derived D3/D4 presented as owner-locked, non-reproducible \$3 gen0 command, `GOLDEN\_VIDEOS` path inconsistency, stale-at-merge A0 status). Author fixed \*\*all 8\*\* in amended `41ce99c` (also added observability/job-failure triage to A13). Verified line-by-line on the new head + full local gates: suite \*\*1595? / 11 skip\*\*, cov \*\*85.20%\*\* (floor 70), `ruff` + `mypy` backend training` clean, `git diff --check` clean, branch = exactly 1 commit over `origin/master e4914d1`. Product/strategy doc ? \*\*merge stays owner-gated\*\* (per working-agreement-merging). After merge, the real alpha gates: A1 (serving endpoint, owner spend/CF), A2 (corpus portability ? next concrete PR per \$8), A10/A11 (ship-gate target + WASB-C3, owner/legal). Known real bug tracked as A6: `backend.eval.ablation`?`ablation\_pdf` circular import (fix = defer `export\_pdf` import into `main()`).

154

155 ---

156

157 The 2026-07-02 health audit remediation is \*\*COMPLETE + ARCHIVED\*\*. After the sibling session drove P0?P28 to done, a follow-up session (2026-07-02 PM) \*\*independently re-verified every P0?P2 fix in `origin/master` code at file:line\*\* (all CONFIRMED, CI green on all 5 heads), then closed the last confirmed-but-untracked residuals and archived the umbrella doc. \*\*No open remediation PRs; all 5 repos have 0 non-default branches, 0 orphaned worktrees, 0 dirty files\*\* (full workflow/worktree/branch cleanup done).

158

159 Archived tracker (was live, now archived):

160

161 `badminton-highlight-indexer\docs\archives\past\_projects\AUDIT\_REPORT\_khelsutra-guru\_2026-07-02-COMPLETED-2026-07-02.md`

162

163 Final repo heads:

164

|     |                                           |
|-----|-------------------------------------------|
| 165 | Repo   Head                               |
| 166 | --- ---:                                  |
| 167 | `badminton-highlight-indexer`   `ac6184b` |
| 168 | `khelsutra`   `1639804`                   |
| 169 | `rally-annotator`   `ef9e088`             |
| 170 | `sports-data-collector`   `bfc014f`       |

```

171 | `sports-obsreport` | `a0c7252` |
172
173 ## Follow-up session residuals (2026-07-02 PM) ? all merged on green
174
175 - `sports-data-collector#61` ? **R1-41**: crawler skips the whole match when any per-set
CSV fails (was silently ingesting partial rally data) + regression test. Suite 294?, cov 81.33%.
176 - `#441` ? **R1-24/R1-30**: `git rm --cached` the 7 tracked `scratch/*.py` (they defied
the `scratch/` ignore-contract; secret-leak vector). Files kept on disk, now ignored.
177 - `#442` ? **R1-11/R1-26**: pin `ruff==0.15.16` / `mypy==2.1.0` in ci + nightly
(verified green at those versions ? zero churn).
178 - `#446` ? archived the umbrella audit doc + repointed all inbound refs
(README/DOC_STATUS/NEXT_STEPS/test). ? gotcha: first commit only carried the `git mv` because
`git add` aborted on the already-moved old path ? a second commit landed the content edits;
link-check green after.
179 - Filed remaining owner-gated/future items as GitHub issues: **443** (CORS default
`['*']`, R1-12) . **444** (corpus Phase-2 upload decision) . **445** (GCS eval/litmus consume
`gs://` by URI). Already-open: #301 (retention automation), #327 (path-jail gs://), #332 (NSFW),
#384 (CI single-source).
180
181 ## Latest Completed Items
182
183 - `#438`: Roora pilot non-video GCS archive complete; raw videos remain excluded.
184 - `sports-data-collector#56`: Roora pilot golden-label fixtures committed with tests.
185 - `sports-data-collector#60`: six owned golden source MP4 MD5/path manifest committed;
uploads remain owner-gated.
186 - `khelesutra#150`: retired-droplet docs refreshed from current master; stale `#141`
closed superseded.
187 - `#439` and `#440`: tracker/docs final reconciliation complete and merged.
188 - Closed superseded stale PRs: `sports-data-collector#54`, `sports-data-collector#55`,
`khelesutra#141`.
189 - Root non-git audit snapshot replaced with a pointer to the live engine tracker.
190
191 ## Verification
192
193 - Engine `#439` and `#440` CI green, including full offline suite with coverage floor,
ruff, mypy, wheel smoke, markdown links, leak/license checks, and cross-OS golden fixtures.
194 - Engine docs local checks: `python -m tools.check_md_links`; `git diff --check`.
195 - Sports-data-collector `#56`: merge-result local full suite `293 passed, 2 skipped`;
coverage `82.02%` against floor `80`; CI green.
196 - Sports-data-collector `#60`: local full suite `290 passed, 2 skipped`; coverage
`82.02%` against floor `80`; CI green.
197 - Khelesutra `#150`: local `site` tests `95 passed`; web typecheck; web coverage gate;
web e2e/contract smoke `5 passed`; CI green.
198 - Prior UI rows include screencast/Playwright evidence under
`C:\Users\avidu\Projects\khelesutra-guru\khelesutra-screencast\_raw`.
199
200 ## Leftovers
201
202 | Area | Leftover | Status |
203 |---|---|---|
204 | Corpus videos | Decide/upload large source videos: full 4K originals vs proxies, and
wider archive scope. | Owner-gated ? **issue #444**. |
205 | GCS eval/litmus | Make eval/litmus consume `gs://` data directly. | Future project ?
issue #445. |
206 | CORS default | `allow_origins=['*']` lets any site script the localhost appliance. |
Security ? **issue #443**. |
207 | Hosted deployment | Enable upload retention in live infra and clean up
Cloudflare/alpha. | Owner infra/timing ? **301** + CF issues. |
208 | Product backlog | Non-audit feature/quality issues. | Normal backlog. |
209
210 ## Resume Rule
211
212 Before any future work: `git pull --ff-only` in the target repo, branch from
`origin/<base>`, re-check branch/HEAD before staging, and stage explicit paths only.
213

```

## 24. user (2026-07-04T11:50:07.982Z)

```
1 ---
2 name: session-handoff
3 description: "Living handoff: current state, next steps, ramp-up kit, and key
decisions."
4 metadata:
5 node_type: memory
6 type: project
7 ---
8
9 # Session handoff ? khelsutra-guru
10
11 > Last refreshed: 2026-07-04 (end of the e2e session). **A1 IS DONE.** All 7 serving PRs
merged
12 > (#465/#467/#468/#469/#470/#474 on owner LGTM + #476 on green CI), the full e2e PASSED
on the final
13 > image, edge-auth is restored, and the runlog + A1?DONE flip is merged (#477, master
`122f174`).
14 > See [[current-state]] for the blow-by-blow.
15
16 ## ? YOU ARE HERE ? A1 COMPLETE; next = alpha launch backlog
17
18 **A1 is DONE and verified** (runlog:
`docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_a1-e2e_2026-07-04.md`).
19 The live stack: CP rev **00007** (edge-auth ON, image digest `31475f27` = master
`3c97117`),
20 L4 job re-pinned to the same build, `api.khelsutra.guru` ? CF Access login (4-email
allowlist),
21 `maxScale=1`. Owner said KEEP it running.
22
23 **Natural next grabs (all ungated unless noted):**
24 1. **A13 alpha launch review** (ALPHA_LAUNCH_READINESS §7) ?
runbook/consent/observability/GO-NO-GO.
25 2. **khelsutra UX backlog** ? the 13-issue triage in [[next-session-ux-localization]]
(async /api/compile
26 #329 kills the 524, NSFW litmus #332, etc.).
27 3. Engine follow-ups filed this session: **471** (per-instance SQLite state ? alpha
runs maxScale=1,
28 durable fix open) . **472** (cert-renewal watch ~2026-09-30) . optional `--revision-
suffix` for
29 `gen_service_yaml.py` (the `services replace` no-op gotcha, runlog §5).
30 4. **Rally finding-B promotion (P2b)** ? the other ? Start-here track, untouched by this
session.
31
32 **FIRST STEPS for the next session:** `git -C C:\Users\avidu\Projects\khelsutra-
guru\badminton-highlight-indexer
33 pull --ff-only`; `gh auth status` (should be `avidullu`); read [[current-state]]'s top
checkpoint.
34
35 ---
36
37 ## (A) PR-REVIEW RESOLUTION ? remaining open PRs
38
39 Repo `Khelsutra/badminton-highlight-indexer`. Reviewer = **avidullu** (the owner).
"LGTM" arrives as a
40 PR comment or a formal approval. Merge convention = **squash** (`gh pr merge <n>
--squash --delete-branch`).
41
42 | PR | branch | fixes | head | review state |
43 |---|---|---|---|---|
44 | ~~#465~~ | ~~`fix/allowed-video-root-uri`~~ | hosted-mode input jail vs gs:// | ? |
MERGED 2026-07-04 09:03Z (owner LGTM + 11/11 CI) |
45 | **467** | `fix/upload-gcs-staging` | closes #461 (upload?gs:// staging) | c9bf37e |
round-2 blocker (input_root-scheme gate) ADDRESSED; awaiting LGTM |
```

```

46 | ##468 | `fix/drain-concurrency` | closes #466 (parallel videos) | 21ee141 | round-2
blocker (claim_due_job lost-race false-empty) ADDRESSED; awaiting LGTM |
47 | ##469 | `fix/highlights-signed-url` | closes #463 (32 MiB download ? signed URL) |
ce450e5 | round-2 blocker (configured signed_url_ttl ignored) ADDRESSED; awaiting LGTM |
48 | ##470 | `fix/dockerfile-auth-extra` | base image M9-ready (auth extra) | c63d122 |
NEW this session; 11/11 green (first push tripped the Dockerfile structural guard ? guard
updated). Once merged, the e2e SKIPS the derived-image step |
49 | ##474 | `feat/cloudrun-service-manifest` | closes #473 (gen_service_yaml.py + tests
+ README §3a) | 9de18d5 | NEW this session; encodes maxScale=1 (#471 opt-1), no CLOUD_RUN_JOB,
explicit --edge-auth. Once merged, the e2e regenerates the YAML from the repo |
50
51 Fix worktrees (uncommitted-free, branches pushed): `wt-drain-concurrency`, `wt-upload-
staging`,
52 `wt-signed-download`, `wt-dockerfile-auth`, `wt-yaml-gen` under
`C:\Users\avidu\Projects\khelesutra-guru`;
53 `wt-465fix` is now merged ? safe to `git worktree remove` all of them after their PRs
merge.
54
55 What was already discussed + resolved (so you're not surprised by the threads):
56 - ##465 ? reviewer flagged the offline suite was red
(`tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri`
57 asserts hosted mode rejects gs://). Fixed in cd4bc73: `validate_input_path` now
takes `allowed_uri_root`
58 (wired from `storage.input_root`); a bucket URI is allowed ONLY under input_root,
arbitrary buckets
59 rejected (tenant isolation). The contract test passes unchanged. Replied on the PR.
60 - ##467 ? reviewer: `storage.put(local_src, staged_uri)` dropped the storage config.
Fixed: pass
61 `config=storage_cfg.model_dump()`. Test asserts the config reaches `put()`. Replied.
62 - ##468 ? reviewer withdrew LGTM (round 1): over-broad concurrency. Re-designed +
pushed (8534077):
63 `orchestration._LOCAL_COMPUTE_LANE` (BoundedSemaphore(1)) around the in-process
segmenter + compile
64 stitch (remote polls don't hold it ? overlap; local/compile serialize); broadened
65 `_cloud_dispatch_possible()` so `compute='cloud'` overrides scale too. I also OFFERED
(in a reply) to
66 make the lane *park* via `JobWaiting`/WAIT_AND_RESUME instead of *block* (starvation-
free) ? do that
67 only if the reviewer asks.
68 - ?? OPEN ? a 2nd blocker the reviewer found (round 2), NOT yet fixed. Address this
FIRST on 468:
69 `claim_due_job()` (`backend/infrastructure/database_jobs.py:357-380`) returns `None`
when a worker
70 LOSES the CAS race (`rowcount != 1`); `_drain_due_jobs()`
(`backend/api/job_worker.py:273-276`)
71 treats `None` as "queue empty" and EXITS. So with `max_workers>1`, workers racing
the same first
72 queued row ? losers return `None` ? their drains stop even though other jobs are
queued (reviewer
73 reproduced: 8 queued + 8 claimers ? only 2/8 claimed ? dispatch capacity wasted).
74 FIX: make `claim_due_job()` RETRY on a lost race ? re-SELECT + re-attempt until
it either claims
75 a row OR a fresh SELECT finds no runnable rows (or return a distinct lost-race
sentinel so
76 `_drain_due_jobs` LOOPS instead of exiting). ADD a regression test: N queued
jobs + N concurrent
77 claimers/drains prove N DISTINCT jobs are claimed with no false-empty. Then FULL
suite + lint + push
78 + reply on 468.
79 - ##469 ? reviewer: mirror `put` + `exists` + `signed_url` dropped the storage
config. Fixed: added
80 `orchestration._storage_settings(cfg)` threaded into all three; extracted
`signed_reel_url` to keep
81 `backend/main.py` under its line budget (P23/P24). Tests assert config reaches
exists/signed_url. Replied.

```

```

82
83 **Watch/merge protocol (repeat until all 4 merged):**
84 1. Poll `gh pr view <n> --json state,reviewDecision,comments,reviews` for each open PR
 (every few min).
85 2. On **new change-request/comment** ? address it (fix + FULL local suite + push +
 reply). ? Run the
86 **whole** `pytest tests/` suite before pushing, not just the changed file ? that's
 how #465's CI-red
87 slipped through the first time (`python -m pytest tests/ -q -p no:cacheprovider`, ~2
 min; + `ruff check`
88 + `mypy backend/...`).
89 3. On **LGTM/approval** for a PR ? verify `gh pr checks <n>` all pass +
 `mergeable=MERGEABLE`, then
90 `gh pr merge <n> --squash --delete-branch`.
91 4. **Conflicts after a merge (expected):** #467 & #469 both touch
 `backend/orchestration.py`; #468 & #469
92 both touch `backend/main.py`. When one merges, re-check the others' `mergeable`; if
 `CONFLICTING`,
93 in a **worktree off origin/master** merge master into the branch, resolve, re-run the
 FULL suite +
94 lint, push. (Git-safety: `git worktree add <path> <branch>`; stage explicit paths;
 re-verify branch
95 before push; never `git add -A`. Shared checkout ? sibling sessions may move master.)
96
97 ---
98
99 ## (B) FULL E2E DEPLOY TEST ? run AFTER all four PRs are merged
100
101 Goal: prove the merged fixes work on a real deployed build ? the >32 MiB signed-URL
 download (#463) and
102 the browser-upload?gs:// staging (#461) ? then restore the live edge-auth state and flip
 A1 ? DONE.
103
104 1. **Rebuild the worker image from merged master** (has #461/#463/#465/#468 backend code
 + #462's
105 `[storage-gcs,cloud-run]`): `gcloud builds submit
 --config=deploy/cloudrun/cloudbuild.yaml` (heavy
106 CUDA build, ~15 min; region asia-southeast1). ? The base Dockerfile installs
 `[storage-gcs,cloud-run]`
107 but NOT the `auth` extra (PyJWT[crypto]) that M9 edge-auth needs ? the *deployed*
 image gets PyJWT via
108 a derived overlay. **Recommended follow-up:** add `auth` to the base Dockerfile so a
 rebuild is
109 M9-ready (a small 6th PR). For the e2e itself, edge-auth is OFF, so PyJWT isn't
 required.
110 2. **Build the derived control-plane image** = base + PyJWT: context `Dockerfile` =
 `FROM
111 asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest`
 +
112 `RUN python3 -m pip install --no-cache-dir "google-cloud-run>=0.10,<1"
 "PyJWT[crypto]>=2.8"; tag
113 `.../rally/rally-control-plane:latest`. `gcloud builds submit <ctx> --tag
 --region asia-southeast1`.
114 (No more paths.py/backend overlays needed once the fixes are IN the base image ? the
 prior session
115 overlaid #465's OLD blanket paths.py into the live image; the rebuild replaces it
 with the corrected code.)
116 3. **Deploy CP with edge-auth OFF** so you can drive it with an identity token (no CF
 login needed for
117 the test). Regenerate the service YAML (config below) with `EDGE_AUTH` unset and
118 `gcloud run services replace <yaml> --region asia-southeast1`. Run gcloud from
 PowerShell.
119 4. **Run the CUJ + verify** (private service ? auth every curl with
120 `-H "Authorization: Bearer $(gcloud auth print-identity-token)"`; URL below):
121 - `POST /api/process {"video_path":"gs://khehsutra-rally-

```

```

serving/videos/MahadevpuraSingles.MP4","segmenter":"wasb_hybrid","compute":"cloud"}`
122 ? poll `/api/jobs/{id}` to `done` (L4 run ~13 min) ? note segment ids.
123 - `POST /api/compile
{"video_id":<id>,"segment_ids":[1..N],"output_filename":"e2e.mp4"}` ? poll to done.
124 - ##463 check: ** `GET /api/highlights/{video_id}/e2e.mp4` should now
307-redirect to a
125 `storage.googleapis.com` signed URL, and `curl -L` should download the FULL >32 MiB
reel (no edge-500).
126 Also confirm the reel is mirrored at `gs://khelsutra-rally-
serving/highlights/<video_id>/e2e.mp4`.
127 - ##461 check: ** upload a small mp4 via `/api/upload/init`?`/part/0`?`/complete`,
then `POST /api/process`
128 with the returned LOCAL path + `compute:cloud` ? it should STAGE to gs:// and
dispatch (not hard-fail
129 with "cloud-serving needs a cloud input"). (Signing was already verified: `gcloud
storage sign-url`
130 impersonating the compute SA produced a working URL ? the SA has
self-`serviceAccountTokenCreator`.
131 5. **Restore edge-auth ON** (regenerate the YAML with `EDGE_AUTH=1` + redeploy) so
api.khelsutra.guru's
132 live posture is back. Verify `/api/process` sans CF JWT ? 401 and
`https://api.khelsutra.guru/api/health`
133 ? 302 to the CF login.
134 6. **Write** `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-
serving_a1-e2e_<date>.md` (a NEW
135 file ? don't clobber the existing #457 or #464 runlogs), update [[current-state]] +
this handoff, and
136 **flip A1 ? DONE** in `docs/ALPHA_LAUNCH_READINESS.md` §7 (all DoD met: endpoint +
health + edge-auth +
137 upload?process?jobs?compile?download).
138
139 **Config for the CP service (base64 into a startup wrapper via
`RALLY_APP_HOME=/tmp/apphome/config.json`):**
140 ```json
141 {"deployment":{"profile":"cloud-
serving","compute_target":"cloud_run"},"indexing":{"skip_ai_handoff":true},
142 "storage":{"backend":"gcs","output_root":"gs://khelsutra-rally-
serving","input_backend":"gcs",
143 "input_root":"gs://khelsutra-rally-serving/videos"},
144 "security":{"edge_auth_enabled":<EDGE_AUTH>,"cf_team_domain":"https://khelsutra.cloudflareaccess.com",
145 "cf_access_aud":"c980da67ed888fff935fcd913ae1eb4da63f9d65805d6e1c5ca76f92f517082b",
146 "allowed_video_root":"/tmp/apphome/output"}}
147 ```
148 CP deploy (per `deploy/cloudrun/README.md` §3a): --command /bin/bash --args` a -c`
wrapper that
149 `printf %s '<b64>' | base64 -d > /tmp/apphome/config.json && export
RALLY_APP_HOME=/tmp/apphome && exec
150 python3 -m backend.main --server --host 0.0.0.0 --port 8080`; flags --no-cpu-throttling
--min-instances 1
151 --cpu 2 --memory 4Gi --port 8080 --timeout 300`; env `DEPLOYMENT_PROFILE=cloud-serving,
152 CLOUD_RUN_REGION=asia-southeast1,GOOGLE_CLOUD_PROJECT=gen-lang-
client-0306026826,RALLY_ALLOWED_HOSTS=``.
153 ? `CLOUD_RUN_JOB` is a RESERVED env name ? DON'T set it (code default `rally-worker` is
correct).
154 Deploy via a `gcloud run services replace <service.yaml>` MANIFEST (shell metachars in
the wrapper break
155 through the cmd-shim otherwise). The prior session's generator `gen_service_yaml.py` +
the derived-image
156 `Dockerfile` lived in the prior scratchpad ? rebuild them from the config + notes above.
157
158 ## Deployed state (live now ? owner said KEEP running)
159 - Project `gen-lang-client-0306026826`, region `asia-southeast1`. Compute SA
160 `492532266377-compute@developer.gserviceaccount.com` (has `roles/editor` +
self-`serviceAccountTokenCreator`).

```

```

161 - **L4 Job** `rally-worker` (nvidia-l4, 8vCPU/32Gi, FUSE `khelsutra-rally-serving`?/gcs,
`WASB_WEIGHTS=/gcs/models/?`).
162 - **CP Service** `rally-control-plane` rev **00004, edge-auth ON**, PRIVATE (IAM +
allUsers-invoker so CF can
163 reach it; app still requires the CF JWT). URL `https://rally-control-plane-2bosgeukpa-
as.a.run.app`.
164 Deployed image = `?/rally/rally-control-plane:latest` = rally-worker + google-cloud-
run + PyJWT + the
165 **OLD blanket #465 paths.py overlay** (the corrected input_root-scoped code is in PR
#465 ? the e2e
166 rebuild replaces it; harmless for the closed 4-email alpha meanwhile).
167 - **M9 LIVE:** `api.khelsutra.guru` ? CNAME `ghs.googlehosted.com` (PROXIED) via Cloud
Run domain-mapping,
168 fronted by the "KhelSutra Studio (alpha)" CF Access app (team
`https://khelsutra.cloudflareaccess.com`,
169 aud `c980da67?082b`, 4-email allowlist:
avi.dullu@sheendullu13@suneetadullu@avidullu.svp@).
170 **Add a tester** = add their email to that app's "alpha allowlist" policy (CF
dashboard or API; token at
171 `G:\My Drive\cloudflare.token.txt`, has DNS + Access-Apps edit but NOT Rulesets/Config
nor
172 `/access/organizations` ? team domain was recovered via the public certs probe). CF
acct
173 `213c12cf60de9041dd8a3521aa206472`, zone `65ef7770a6f25697843b53a093504184`. ? Google-
managed cert
174 auto-renew (~90d) can stall behind CF proxy ? switch to a CF Origin CA cert if it
does.
175
176 ## Gotchas (learned the hard way)
177 - Run **gcloud from PowerShell**, not git-bash (MSYS mangles `/gcs`, `/tmp`); quote
comma-dict flags
178 (`--add-volume "name=?,type=?"`) or the cmd-shim splits them.
179 - The CP uses per-instance **ephemeral SQLite + /tmp** ? DB/reel don't survive a
redploy or a second
180 instance. That's WHY #463 mirrors reels to gs:// + #461 restages uploads. maxScale is
3.
181 - **Pre-merge:** run the WHOLE suite locally, not per-file (the #465 CI-red lesson).
182
183 ## Key decisions (don't relitigate)
184 - WASB-only alpha (`skip_ai_handoff=true`); config.json wins over the cloud-serving
preset (loader.py:194).
185 - Reuse the worker image as the CP (+ google-cloud-run + PyJWT); IAM-private + CF Access
is the tester auth.
186 - #468 concurrency: serialize local/compile via `_LOCAL_COMPUTE_LANE`, parallelize only
remote polls; the
187 *park*-instead-of-*block* refinement is offered to the reviewer (do it only if they
ask).
188 - A1 stays NOT-DONE until the 4 PRs merge + the e2e passes; then flip it.
189
190 ## Ramp-up kit
191 - The 4 PRs (links via `gh pr list`), their review threads, and CI. Issues:
#461/#463/#466 (fixed by the PRs).
192 - Runlog already in repo: `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-
serving_a1-cuj_2026-07-04.md`
193 (PR #464, merged) ? the full A1 deploy story.
194 - Code: `deploy/cloudrun/README.md` $2/$3/$3a + `Dockerfile` · `backend/main.py`
(routes; `/api/highlights`)
195 · `backend/orchestration.py`
(`_maybe_dispatch_remote`, `_execute_compile`, `signed_reel_url`, `_LOCAL_COMPUTE_LANE`)
196 · `backend/api/job_worker.py`
(`configure_drain`, `_resolve_drain_workers`, `_cloud_dispatch_possible`)
197 · `backend/storage/` (`put`/`exists`/`signed_url`; gcs.py keyless signBlob) ·
`backend/config/models.py`
198 (`deployment.max_concurrent_remote_jobs`) · `backend/utils/paths.py`
(`validate_input_path` allowed_uri_root).

```

```
199 - Repo checkout: `C:\Users\avidu\Projects\kheIsutra-guru\badminton-highlight-indexer`
(master @ 4847232 at handoff).
200
```

## 25. user (2026-07-04T11:50:31.925Z)

```
1 ---
2 name: feedback-tail-long-job-logs
3 description: "User wants progress logs/summaries tailed to them during long-running
background jobs, not just the final result"
4 metadata:
5 node_type: memory
6 type: feedback
7 originSessionId: 42211f80-2169-4025-ad9d-99c1aa29ff70
8 ---
9
10 For any long-running background job (e.g. the WASB GPU inference pass, bulk fetches,
11 big test runs), the user wants **periodic progress surfaced to them while it runs** ?
12 a tail of the log lines plus a short summary (% done, rate, ETA, and resource use like
13 GPU util/mem). This is a "dear ask": don't just report the final result.
14
15 **Why:** visibility into long jobs ? the user wants to see it's alive and progressing,
16 catch a stall/regression early, and have an anchor on how long to wait.
17
18 **? CADENCE REVERSED 2026-06-16 ? PROACTIVE for long jobs now.** Originally the user
wanted
19 ON-DEMAND only (a "tail" trigger; keep quiet while away; declined `/loop`). That stance
20 **FAILED**: a detached GPU extraction died silently ~20s in and sat dead for **~6.5
hours**
21 while they were away, unnoticed, wasting free GPU time. After that the user gave a
STANDING
22 RULE: **"always check the status of any background task that runs for more than 20
minutes,
23 and log it every 5 minutes from then on."** So:
24 - **Long jobs (running >20 min): set up a PROACTIVE heartbeat** ? a `Monitor` (or
25 run_in_background poll loop) that emits a status line **every 5 minutes** with
reels/items
26 done, GPU mem+util, and the last log line, AND **breaks/alerts immediately on stall or
27 failure** (so another silent death is caught fast). A frozen "last log line" + idle
GPU in
28 the heartbeat = stall ? investigate. This is now the DEFAULT, not opt-in.
29 - **The on-demand "tail" trigger still works** for an immediate snapshot anytime (it's
30 additive, not replaced).
31
32 **Why:** visibility into long jobs ? see it's alive + progressing, catch a stall/death
EARLY
33 (the 6.5h silent death is the cautionary tale), and have an anchor on how long to wait.
34
35 **How to apply:**
36 - For any background job expected to exceed ~20 min, arm a 5-min heartbeat monitor at
launch.
37 - Treat a short message like "tail" as a request for an immediate progress snapshot too.
38 - A good snapshot/heartbeat = last few log lines + (windows/items done / total, rate,
ETA) +
39 GPU/CPU/mem. For WASB also show cached-windows / manifest windows_done.
40 - Robust launch for long GPU jobs = a detached **Scheduled Task** + in-process keep-
awake
41 (`SetThreadExecutionState`, beats Modern-Standby S0 which ignores powercfg timeouts) +
42 self-restart loop (pattern: `scratch/extract_val_until_done.ps1` / the old
`FusionGoldenExtract`).
43 A plain harness background job does NOT survive context/console boundaries ? see
[[gotcha-long-runs-gpu-wsl]].
44 - Job logs live at the background task's output file (the path returned at launch).
45 - **Reusable heartbeat = `scratch/watchdog_gx128.py`** (generalize the paths): a poll
loop that
```



```

46 EXITS (? harness re-notifies the agent) on the FIRST of DONE / STALL (windows_done
frozen >180s)
47 / HEARTBEAT (~5 min) ? turning a silent stall into a fast ping. Re-arm after each
heartbeat exit.
48
49 *** VIOLATED A THIRD TIME 2026-07-04 (first CLOUD run ? A1 litmus on Cloud Run L4).**
Dispatched the
50 GPU job with only a terminal-state monitor, no live log tail; the owner had to prod
("any long running
51 job you must ensure is tailed by you!"). The rule applies UNCHANGED to CLOUD jobs: **at
dispatch, arm
52 BOTH (a) a terminal-state monitor AND (b) a Cloud Logging tail** (poll `gcloud logging
read` ~60s,
53 grep progress + failure signatures `worker|wasb|rally|ERROR|Traceback|CUDA|OOM|Killed`,
comm-dedup).
54 Cloud-specific stall causes to check when "nothing is happening": (1) **gcloud
interactive prompts
55 hang silently non-interactively** ? `run jobs create --gpu` prompts for zonal
redundancy; empty output
56 + nothing created ? suspect a prompt; pass `--no-gpu-zonal-redundancy` explicitly (also
half GPU price
57 for batch). (2) First execution of a big CUDA image sits in "Container image import in
progress" for
58 minutes ? that's progress, not a stall; it shows in `executions describe
status.conditions`, NOT logs.
59
60 *** VIOLATED AGAIN 2026-06-19 (GX010128 WASB run).** Launched the GPU run with NO
heartbeat, leaned
61 on the harness "notify on completion" ? but a **sleep-kill fires no clean completion**,
so the stall
62 sat unseen ~49 min (caught only when the owner asked "update?"). Compounded it by
asserting the
63 resumable cache "will resume" without verifying ? the relaunch actually **reset to 0 and
lost 21k
64 windows**. Lesson (re-burned): (1) arm the watchdog AT LAUNCH for any >20-min job, never
rely on the
65 completion notification alone; (2) after a resume, VERIFY windows_done jumped to the
cached value ?
66 don't assert it; (3) pair with keep-awake (`scratch/keepawake.py`) so it isn't sleep-
killed. See
67 [[gotcha-long-runs-gpu-wsl]].
68

```

## 26. user (2026-07-04T11:50:33.100Z)

```

C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\orchestration.py:704:
and survive the ``stitching.min_shot_count`` floor (segments WITHOUT a known shot_count are
exempt
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\eval\ablation.py:34:
- **structural signals are exempt from auto-demotion** ? a guard whose value is in failure modes

```

## 27. user (2026-07-04T11:50:43.016Z)

```

C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\main.py:51: auth
as edge_auth, # M9: Cloudflare Access edge-auth (default-OFF)
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\main.py:678:
edge_auth.resolve_tenant(request.headers, request.app.state.config)
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\main.py:679:
except edge_auth.EdgeAuthError as e:
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\main.py:750:
owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\main.py:751:
except edge_auth.EdgeAuthError as e:
C:\Users\avidu\Projects\khehsutra-guru\badminton-highlight-indexer\backend\main.py:908: #
unchanged). When edge_auth_enabled, the Cf-Access JWT is required + verified (a missing or

```

```

C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\main.py:911:
owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\main.py:912:
except edge_auth.EdgeAuthError as e:
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\main.py:1108:
owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\main.py:1109:
except edge_auth.EdgeAuthError as e:
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\main.py:1305:
"security.edge_auth_enabled. Expose on a network ONLY behind an auth gate.",
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\main.py:1624:
f"gate (edge_auth_enabled / a reverse proxy) is in front, and set "
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\server_runtime.py:50:
when an edge-auth profile is active (`security.edge_auth_enabled` ? i.e. there IS an auth gate
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\server_runtime.py:59:
if getattr(sec, "edge_auth_enabled", False):
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\api\auth.py:3:**DEFAULT-OFF:** when `security.edge_auth_enabled` is False the
resolver is a strict NO-OP (returns
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\api\auth.py:5:`Cf-
Access-Jwt-Assertion` header is **required** and **verified** (RS256) against the team JWKS ?
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\api\auth.py:23:CF_ACCESS_HEADER = "Cf-Access-Jwt-Assertion"
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\api\auth.py:82:
"(pip install -e .[auth]) on a host with edge_auth_enabled."
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\api\auth.py:121:
DEFAULT-OFF (`security.edge_auth_enabled=False`) ? `None` (the local single-user;
unchanged).
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\api\auth.py:126:
if not sec or not getattr(sec, "edge_auth_enabled", False):
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\api\auth.py:135:
"edge_auth_enabled but security.cf_team_domain / cf_access_aud are not configured"
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\models.py:375:
DEFAULT-OFF: edge_auth_enabled=False ? no JWT is required/verified (the local single-user
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\models.py:376:
path is unchanged). When True, /api/process requires + verifies the `Cf-Access-Jwt-Assertion`
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\models.py:381:
edge_auth_enabled: bool = False
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\startup.py:19:behind an edge gate (the owner flips
`security.edge_auth_enabled` on for the Cloudflare-Access
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\startup.py:30:a strict **no-op** ? exactly the pre-M5 behaviour; with
`edge_auth_enabled=False` (today's default)
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\startup.py:94:
"""Exposure-safety posture for an *exposed* HTTP server. Gated on `security.edge_auth_enabled`
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\startup.py:101: if not sec or not getattr(sec, "edge_auth_enabled",
False):
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\startup.py:115: "security.edge_auth_enabled is ON (engine
exposed) but security.allowed_video_root is "
C:\Users\avidu\Projects\khealsutra-guru\badminton-highlight-indexer\backend\config\startup.py:134: f"security.edge_auth_enabled is ON but
{missing} is not configured ? the Cf-Access JWT "

```

[Showing results with pagination = limit: 30]

## 28. assistant (2026-07-04T11:50:59.910Z)

One quick check ? whether `/api/capabilities` is behind the JWT gate.

## 29. user (2026-07-04T11:51:01.580Z)

321:@app.get("/api/health")

```

322: def health(request: Request):
323- """Cheap liveness/readiness probe (no auth) for the SPA's offline banner
324- (HOSTING_PLAN.md:42). Reports the configured sport + default segmenter so the UI can
325- confirm the engine is reachable and which detector is wired. Never raises."""
326- cfg = request.app.state.config
327- sport = getattr(cfg, "sport", None)
328- indexing = getattr(cfg, "indexing", None)
329- --
359: @app.get("/api/capabilities")
360: def capabilities(request: Request):
361- """Honest probe of what THIS box can do ? the first-run wizard renders the compute step
362- from
363- it (PACKAGING_PLAN.md P1). No secrets: reports only whether a Gemini key is PRESENT,
364- never its
365- value. Best-effort + never raises (the wizard degrades gracefully)."""
366- cfg = request.app.state.config
367- indexing = getattr(cfg, "indexing", None)
368- deployment = getattr(cfg, "deployment", None)

```

### 30. assistant (2026-07-04T11:52:01.437Z)

I have everything verified. Let me compose the section.

### 31. user (2026-07-04T11:53:03.095Z)

Structured output provided successfully